

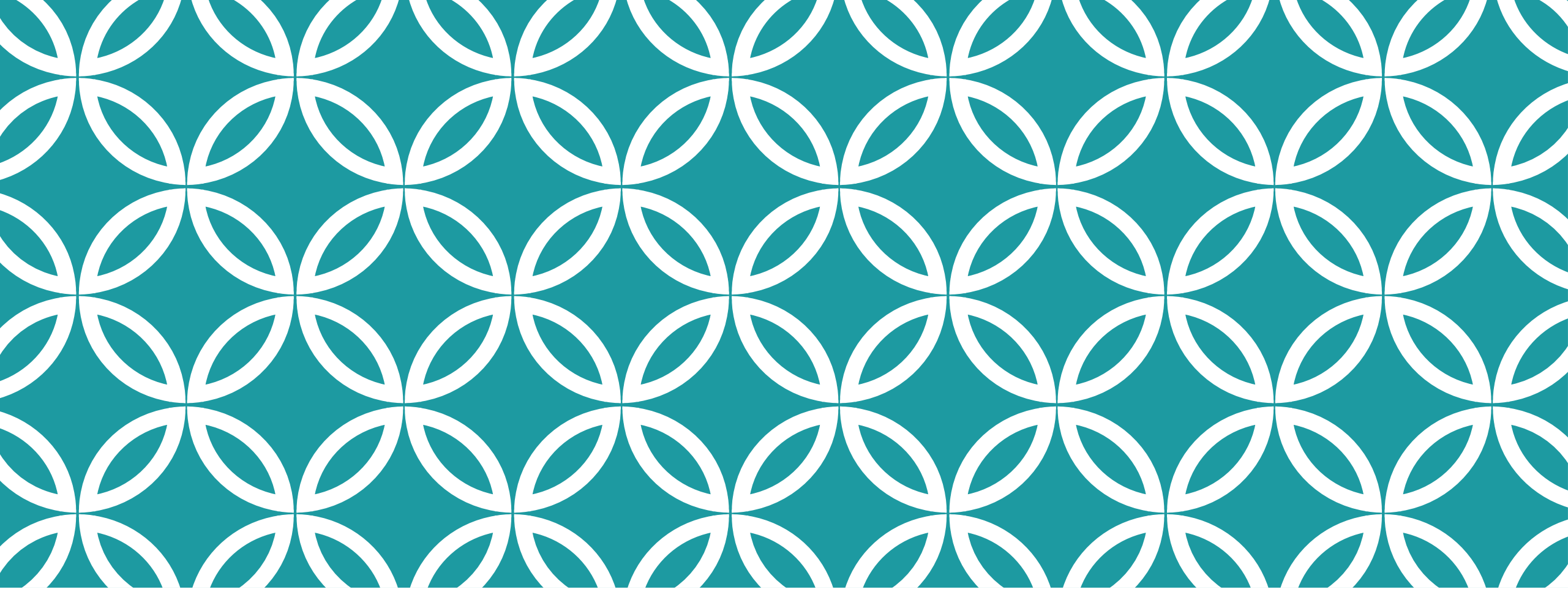


FLAME
UNIVERSITY

PUBP 233 INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

**Week 8: Demography and
Public Health**

Shivakumar Jolad



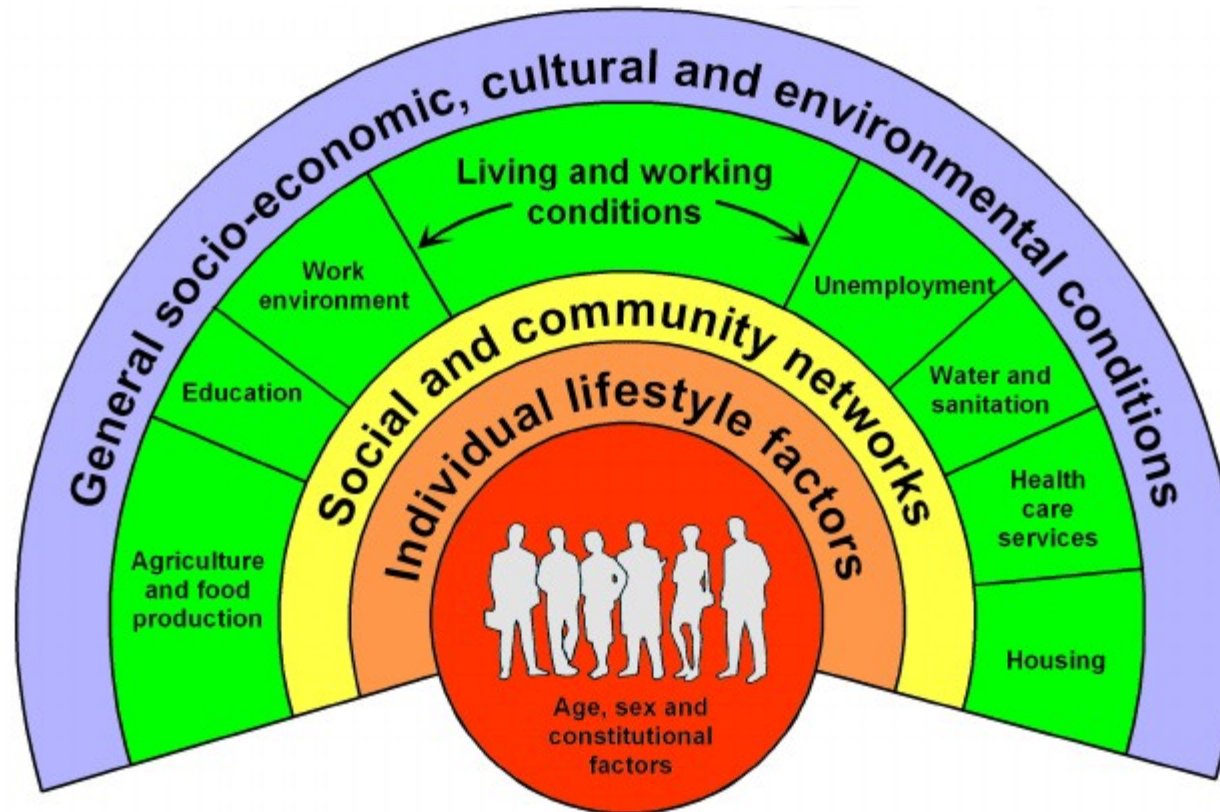
DETERMINANTS OF HEALTH |

SOCIAL DETERMINANTS OF HEALTH

SOCIAL DETERMINANTS OF HEALTH



<https://www.youtube.com/watch?v=8PH4lYfF4Ns>



Source: Dahlgren and Whitehead, 1991

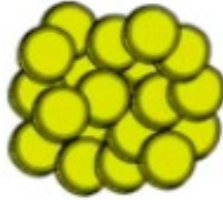
Bambra, Clare & Smith, Katherine & Kennedy, Lynne. (2011). **Politics and health..**

https://www.researchgate.net/figure/Rainbow-model-of-the-determinants-of-health_fig1_49951184

SOCIAL DETERMINANTS OF HEALTH



networks



socio-economic



cultural



environmental



health systems

conditions in which people are
born, grow, live, work
and age

social determinants of
health



smoking



physical
activity



alcohol



diet

What they do



Who they are

age

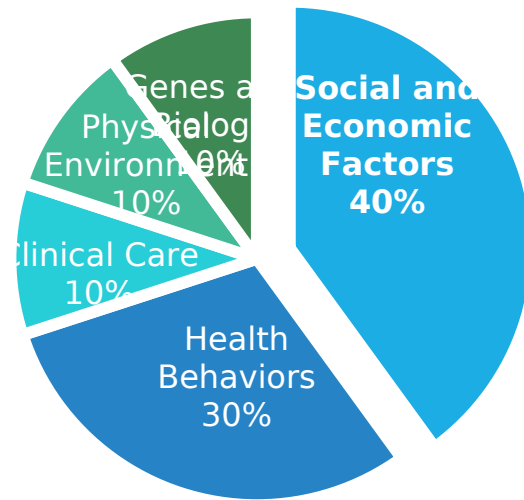
sex

genetic factors

WHAT CREATES HEALTH

Necessary conditions for health (WHO)

Determinants of Health

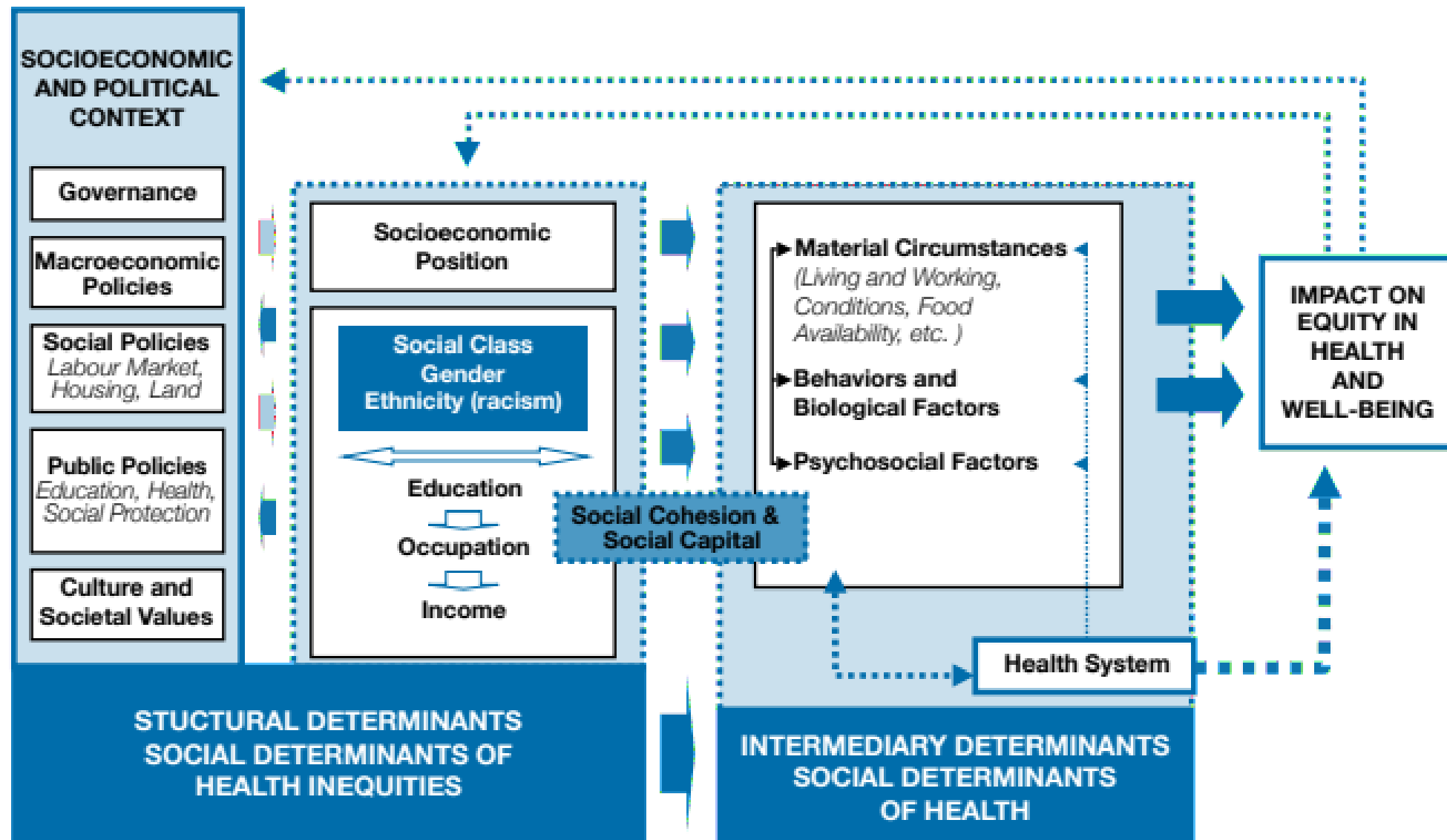


- ☀ Peace
- ☀ Shelter
- ☀ Education
- ☀ Food
- ☀ Income
- ☀ Stable eco-system
- ☀ Sustainable resources
- ☀ Mobility
- ☀ Social justice and equity

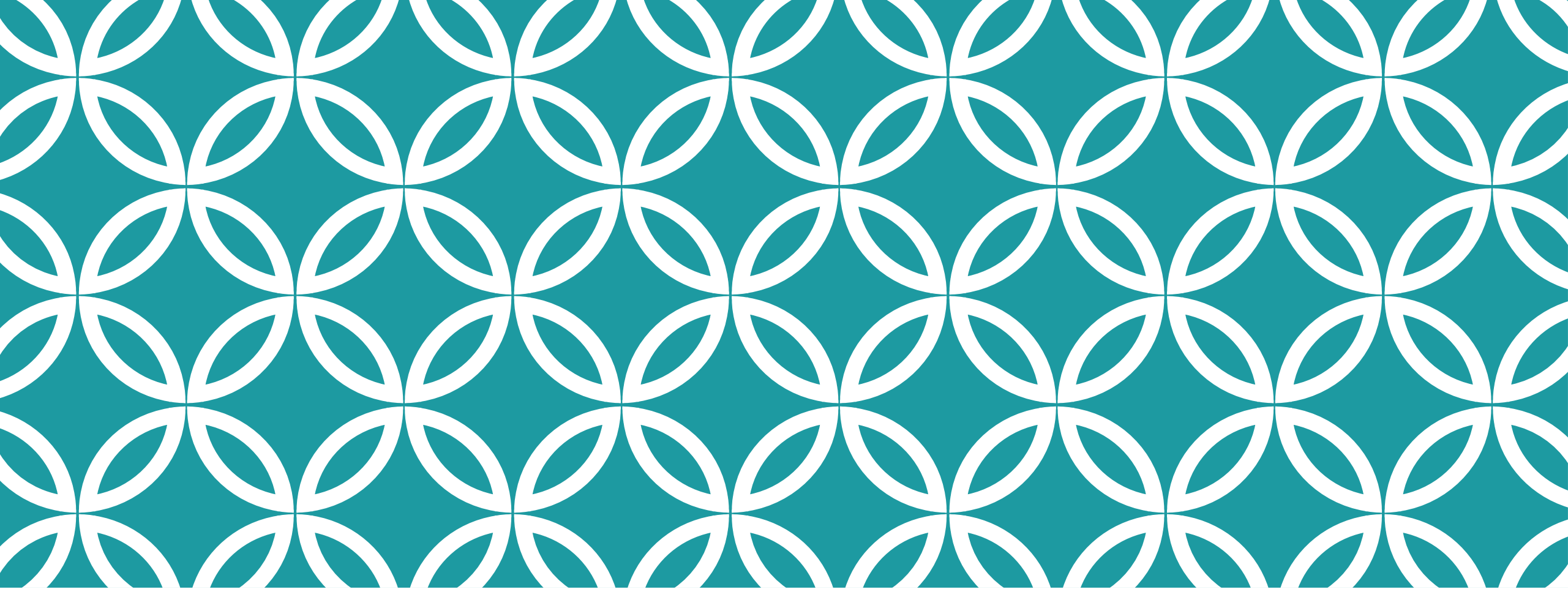
Determinants of Health Model based on frameworks developed by: Tarlov AR. *Ann N Y Acad Sci* 1999; 896: 281-93; and Kindig D, Asada Y, Booske B. *JAMA* 2008; 299(17): 2081-2083.

World Health Organization. Ottawa charter for health promotion. International Conference on Health Promotion: The Move Towards a New Public Health, November 17-21, 1986
Ottawa, Ontario, Canada, 1986. Accessed July 12, 2002 at <<http://www.who.int/hpr/archive/docs/ottawa.html>>.

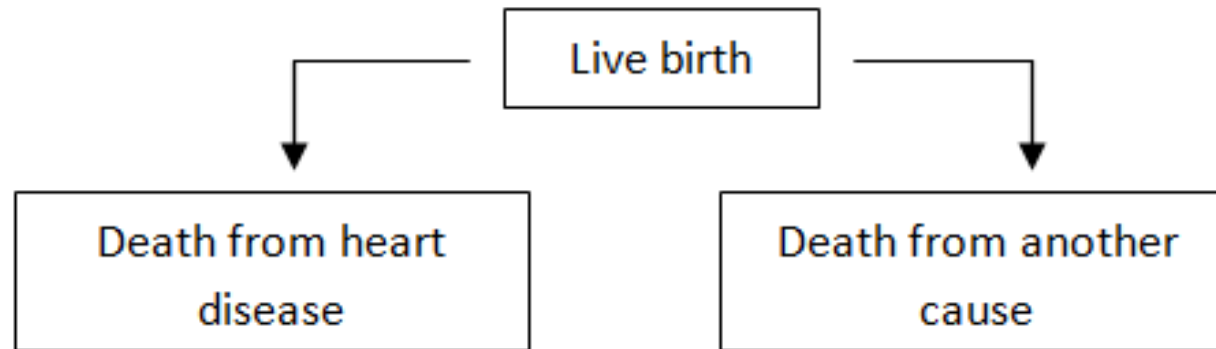
Edward P. Ehlinger, *ADVANCING HEALTH EQUITY: POLICY AND PROGRAM STRATEGIES TO ADDRESS SOCIAL DETERMINANTS OF HEALTH*



Solar O, Irwin A. **A conceptual framework for action on the social determinants of health. Social Determinants of Health** Discussion Paper 2 (Policy and Practice). WHO, 2010



MORTALITY AND PUBLIC HEALTH |



${}_n d_x^i$: life table decrements from cause i in the age interval x to $x+n$:

${}_n d_x^{-i}$: life table decrements from causes other than i in the age interval x to $x+n$:

[http://papp.iussp.org/sessions/papp104_s03/
PAPP104_s03_030_010.html](http://papp.iussp.org/sessions/papp104_s03/PAPP104_s03_030_010.html)

NOTATION

$d(x, n)^i$ = deaths due to cause i , ages x to $x + n$

$q(x, n)^i = \frac{d(x, n)^i}{l(x)}$ = probability of deaths due to cause i between ages x to $x + n$

$\sum_i {}_n d_x^i = {}_n d_x$ Sum of deaths due to all causes

$$\sum_i {}_n q_x^i = \sum_i \frac{{}_n d_x^i}{l_x} = \frac{{}_n d_x}{l_x} = {}_n q_x \text{ and } \sum_i {}_n m_x^i = \sum_i \frac{{}_n d_x^i}{{}_n L_x} = \frac{{}_n d_x}{{}_n L_x} = {}_n m_x$$

$${}_n q_x^i = \frac{{}_n \cdot {}_n m_x^i}{1 + ({}_n - {}_n a_x) \cdot {}_n m_x}$$

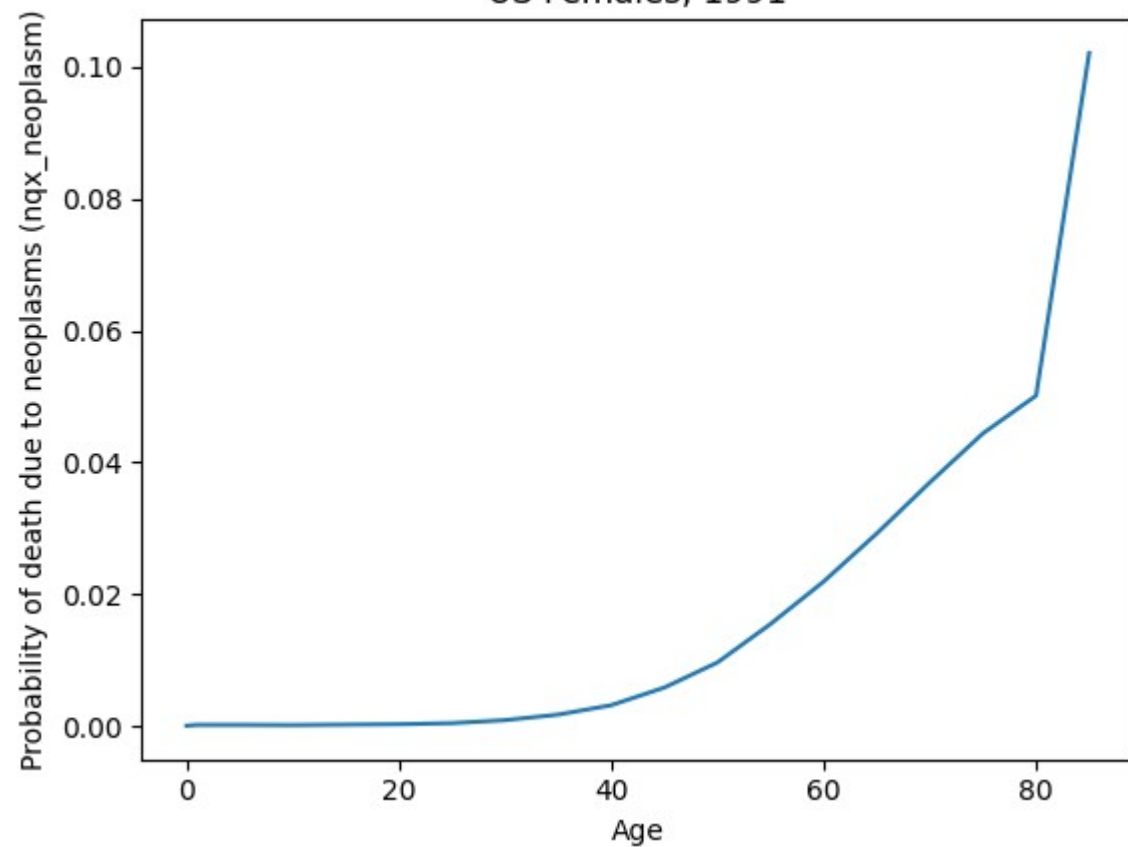
Mistake in Equation

Example: US, females, 1991. Cause i = death from neoplasms, l_x = "master" US life table for 1991, females, for all causes combined

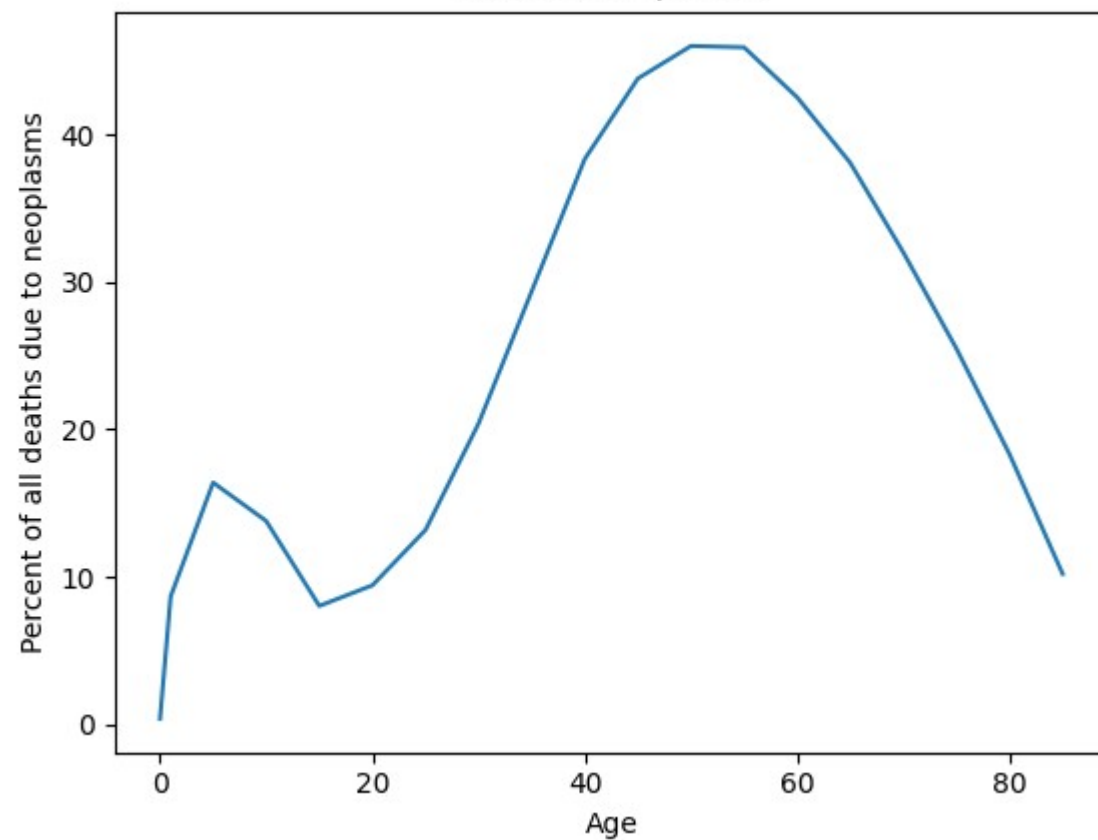
Age x	${}_nD_x$ All deaths	${}_nD_x^i$ Deaths from neoplasms	l_x	${}_nq_x$	${}_nq_x^i$	${}_nd_x^i$	l_x^i
0	15,758	63	100,000	0.00783	0.00003	3	21,205
1	3,169	275	99,217	0.00168	0.00015	14	21,201
5	1,634	268	99,050	0.00092	0.00015	15	21,187
10	1,573	217	98,959	0.00090	0.00012	12	21,172
15	3,955	318	98,870	0.00236	0.00019	19	21,160
20	4,948	467	98,637	0.00262	0.00025	24	21,141
25	6,491	856	98,379	0.00314	0.00041	41	21,117
30	9,428	1,924	98,070	0.00425	0.00087	85	21,076
35	12,027	3,532	97,653	0.00584	0.00171	167	20,991
40	15,543	5,958	97,083	0.00818	0.00314	304	20,823
45	19,264	8,434	96,289	0.01330	0.00582	561	20,519
50	25,384	11,673	95,008	0.02095	0.00963	915	19,958
55	37,211	17,078	93,018	0.03371	0.01547	1,439	19,043
60	59,431	25,263	89,882	0.05155	0.02191	1,969	17,604
65	88,087	33,534	85,249	0.07669	0.02920	2,489	15,634
70	114,693	36,695	78,711	0.11552	0.03696	2,909	13,145
75	143,554	36,571	69,618	0.17427	0.04439	3,091	10,236
80	164,986	30,220	57,486	0.27363	0.05012	2,881	7,146
85	320,578	32,739	41,756	1.00000	0.10212	4,264	4,264
All	1,047,714	246,085				21,205	

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: measuring and modeling population processes*.

Age-specific Probability of Death due to Neoplasms
US Females, 1991



Share of Deaths due to Neoplasms by Age
US Females, 1991



BURDEN OF DISEASE

Attribute mortality and disability to their biological causes and to risk factors for disease.

Global Burden of Disease: “the Global Burden of Diseases, Injuries, and Risk Factors (GBD) enterprise is a systematic, scientific effort to **quantify the comparative magnitude of health loss due to diseases, injuries, and risk factors by age, sex, and geography for specific points in time**”. - Murray *et al.* (2012)

GBD Impetus: study was the proliferation of competing and incompatible estimates of the health impact of different diseases that impeded the setting of health priorities

Objectives:

- ▢ to both **death and disability in an integrated way**
- ▢ to the entire range of diseases and injuries, not just one or a limited set of health conditions
- ▢ in entire populations, at all ages, and of both sexes.

http://papp.iussp.org/sessions/papp104_s05/PAPP104_s05_020_010.html



DISEASE BURDEN: DALY

DALYs for a disease or health condition are calculated as the sum of the Years of Life Lost (YLL) due to premature mortality in the population and the Years Lost due to Disability (YLD) for people living with the health condition or its consequences - WHO.

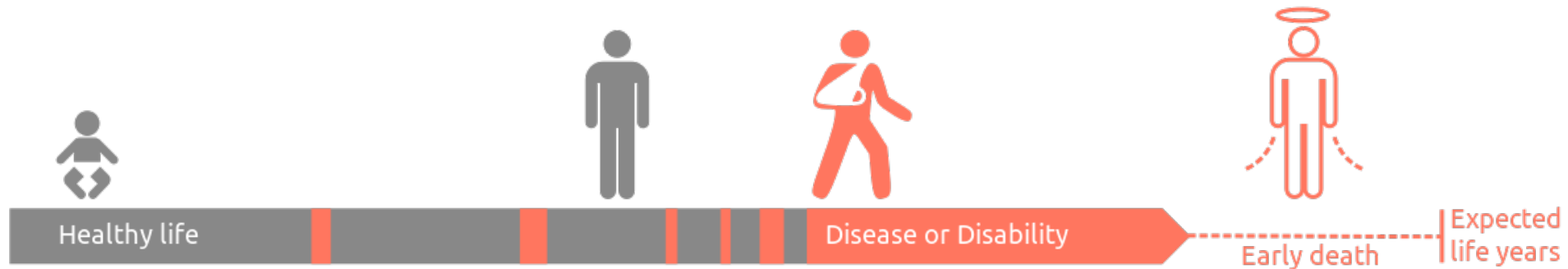
The Years Lived with Disability are weighted by the severity of that disability.

DALY

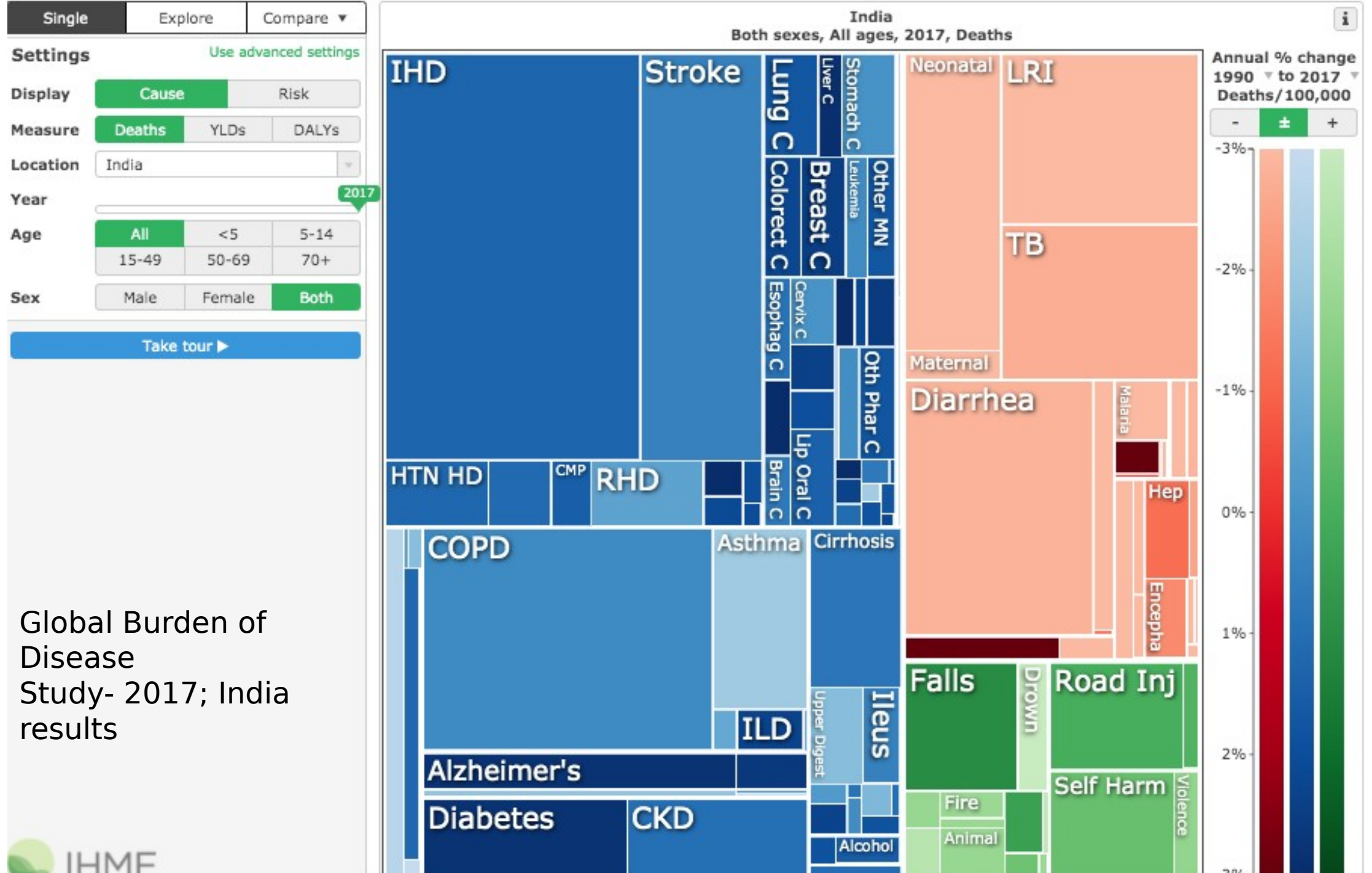
Disability Adjusted Life Years is a measure of overall disease burden, expressed as the cumulative number of years lost due to ill-health, disability or early death

$$= \text{YLD} + \text{YLL}$$

Years Lived with Disability Years of Life Lost



http://en.wikipedia.org/wiki/Disability-adjusted_life_year



CALCULATING DALY

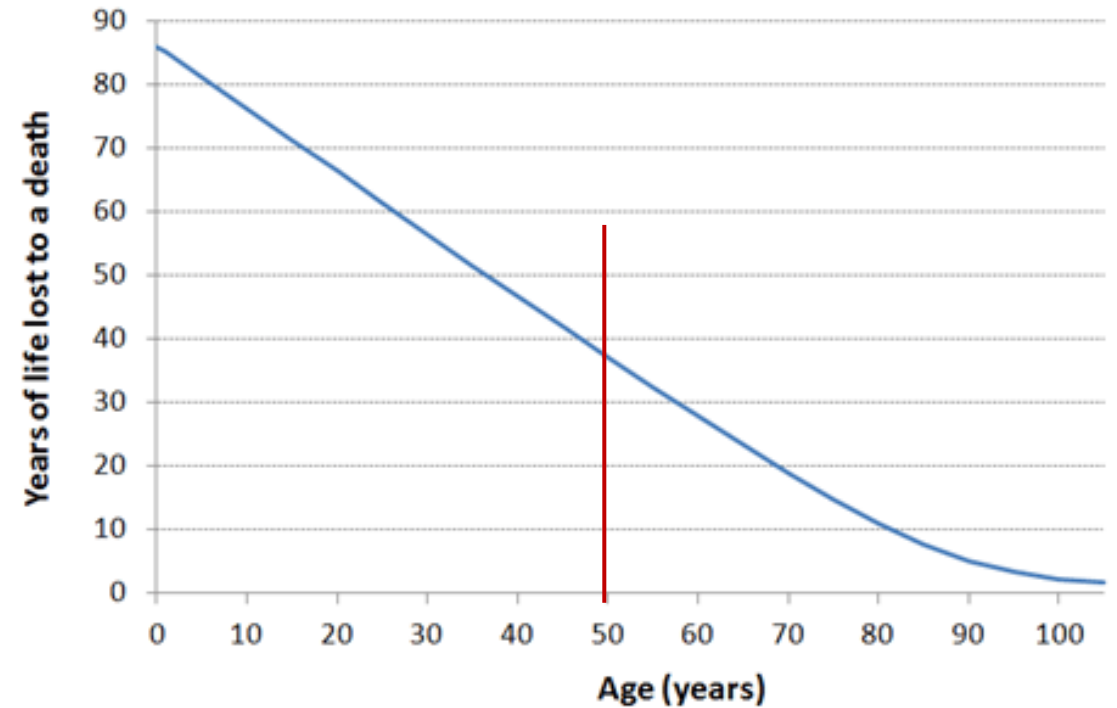
YLD = Number of cases × duration till remission or death
× disability weight (1)

YLL = Number of deaths
× life expectancy at the age of death (2)

DALY = YLD + YLL (3)

$$YLD_{i,x} = \sum_{h=i}^{220} W_h \cdot p_{h,i,x} \cdot N_x$$

$$YLL_i = \sum_{x=0}^{\infty} D_{i,x} \cdot e_x^*$$



YLD associated with each disease and injury, i , in each age group, x , are calculated by multiplying the disability weight for each of the **220 health states, W_h** , by the prevalence, **$p_{h,i,x}$** , of that health state as a sequela of that particular disease and the population at risk in the age group, N_x , and:

Devleesschauwer et al. (2014). DALY calculation in practice: a stepwise approach.

International journal of public health, 59(3), 571.

COMPONENTS OF THE GBD STUDY

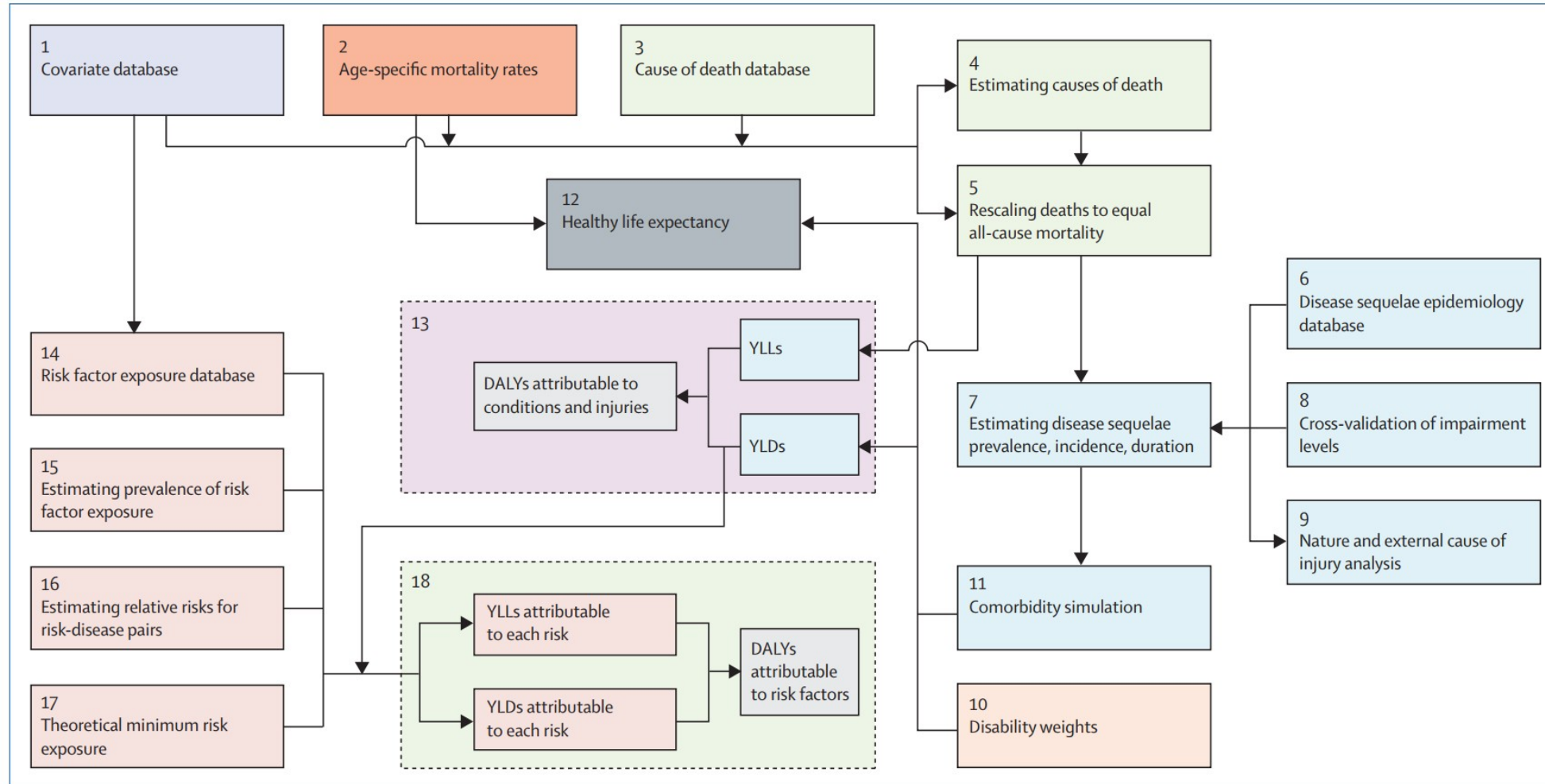


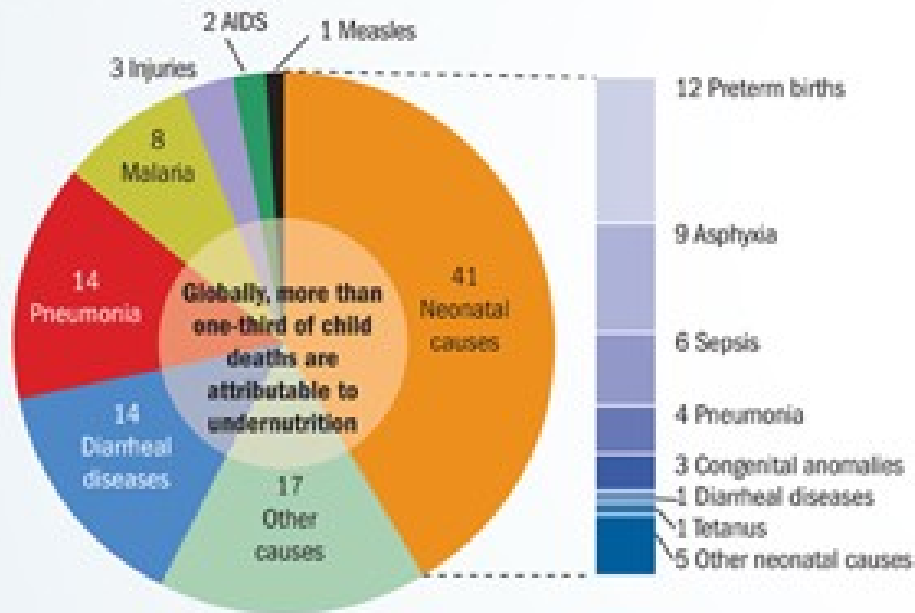
Figure 1: 18 components of the GBD 2010 and their inter-relationships

GBD 2010=Global Burden of Disease Study 2010. DALYs=disability-adjusted life years. YLLs=years of life lost due to premature mortality. YLDs=years lived with disability.

MAJOR CAUSES OF DEATH- CHILDREN AND OLD AGE

Undernutrition and infectious diseases are the major causes of child deaths.

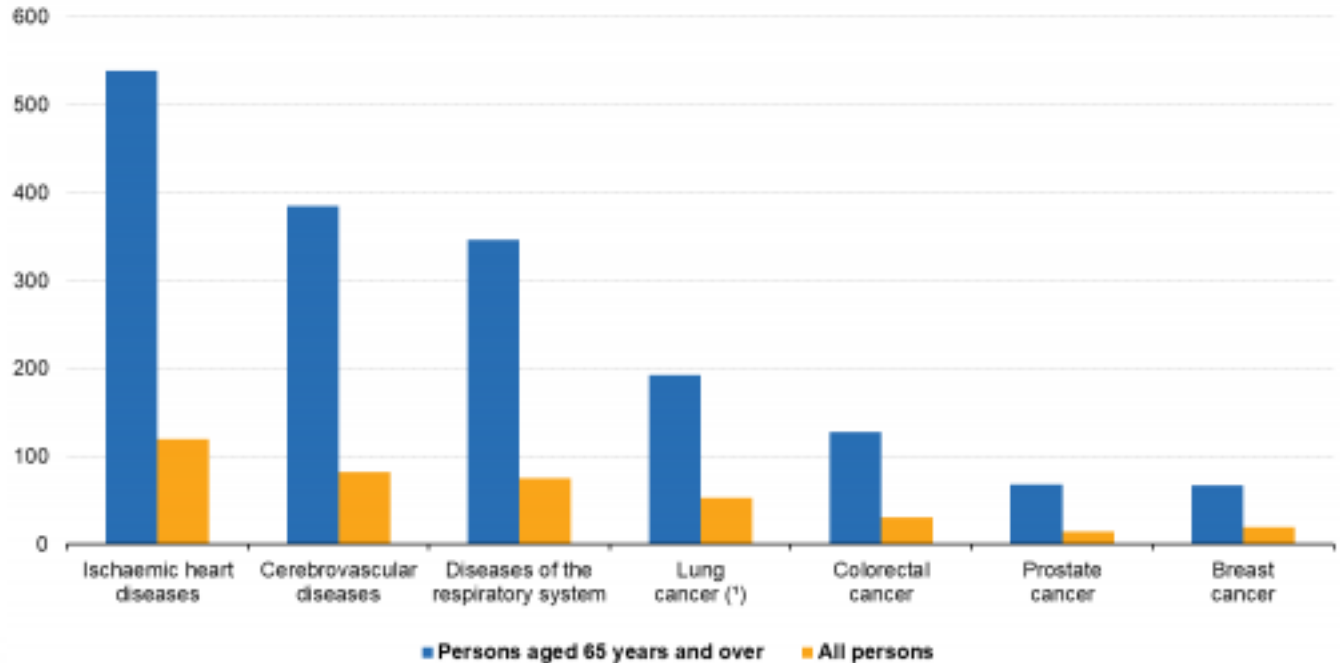
(percent, 2008)



Source: United Nations, The Millennium Development Goals Report 2010.

Major causes of death, EU, 2016

(standardised death rates per 100 000 inhabitants)



Note: the three most common causes of death among persons aged 65 years and over are diseases of the circulatory system, cancer and diseases of the respiratory system.

(*) Trachea, bronchus and lung.

Source: Eurostat (online data code: hlth_cd_asdr2)

eurostat

<https://www.imf.org/external/pubs/ft/fandd/2010/09/picture.htm>

Major causes of death, EU, 2016
(standardised death rates per 100 000 inhabitants)
Source: Eurostat ([hlth_cd_asdr2](#))

DALY IN INDIA

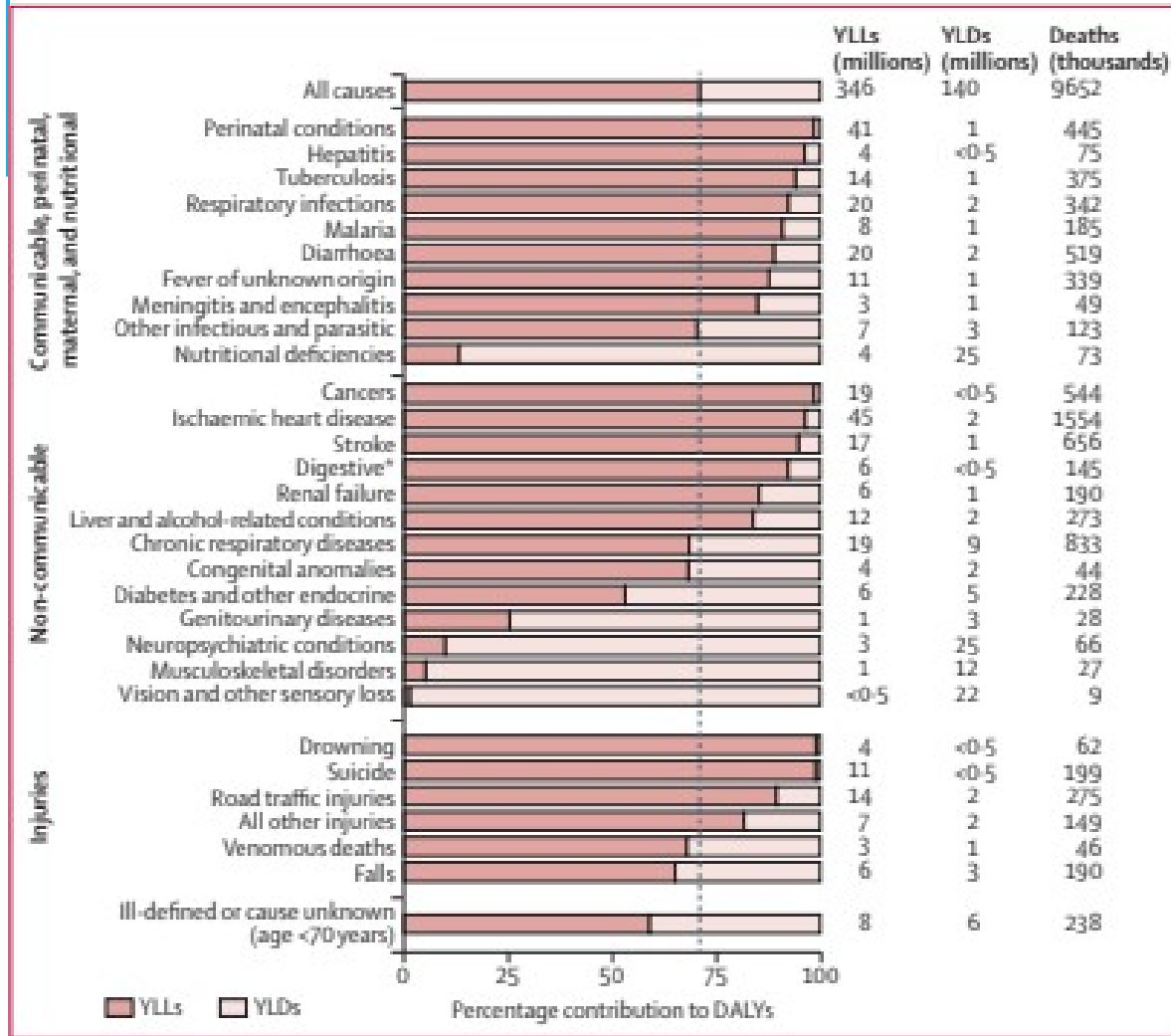


Figure 2: Contribution of YLLs and YLDs for selected major causes of death in India at all ages, 2017
Sexually transmitted infections, selected vaccine-preventable diseases, maternal conditions, epilepsy, rheumatic heart diseases, gastro-oesophageal diseases, and interpersonal violence resulted in a total of 181 000 deaths, with total DALYs comprised of 81% YLLs and 19% YLDs. YLLs=years of life lost. YLDs=years lived with disability. DALY=disability-adjusted life-year. *Digestive excludes gastro-oesophageal diseases and liver and alcohol-related conditions.

India's population 136 crores -- Daily deaths: **23,000 to 26,575 deaths daily**

National Burden Estimates of healthy life lost in India, 2017: **In 2017, there were about 9.7 million deaths and 486 million DALYs in India.-- CDR : 7.2 / 1000**

The ratio of DALYs to deaths was about 50 to one.

More than three quarters of deaths and DALYs occurred in rural areas, and males accounted for 54.3% of all DALYs.

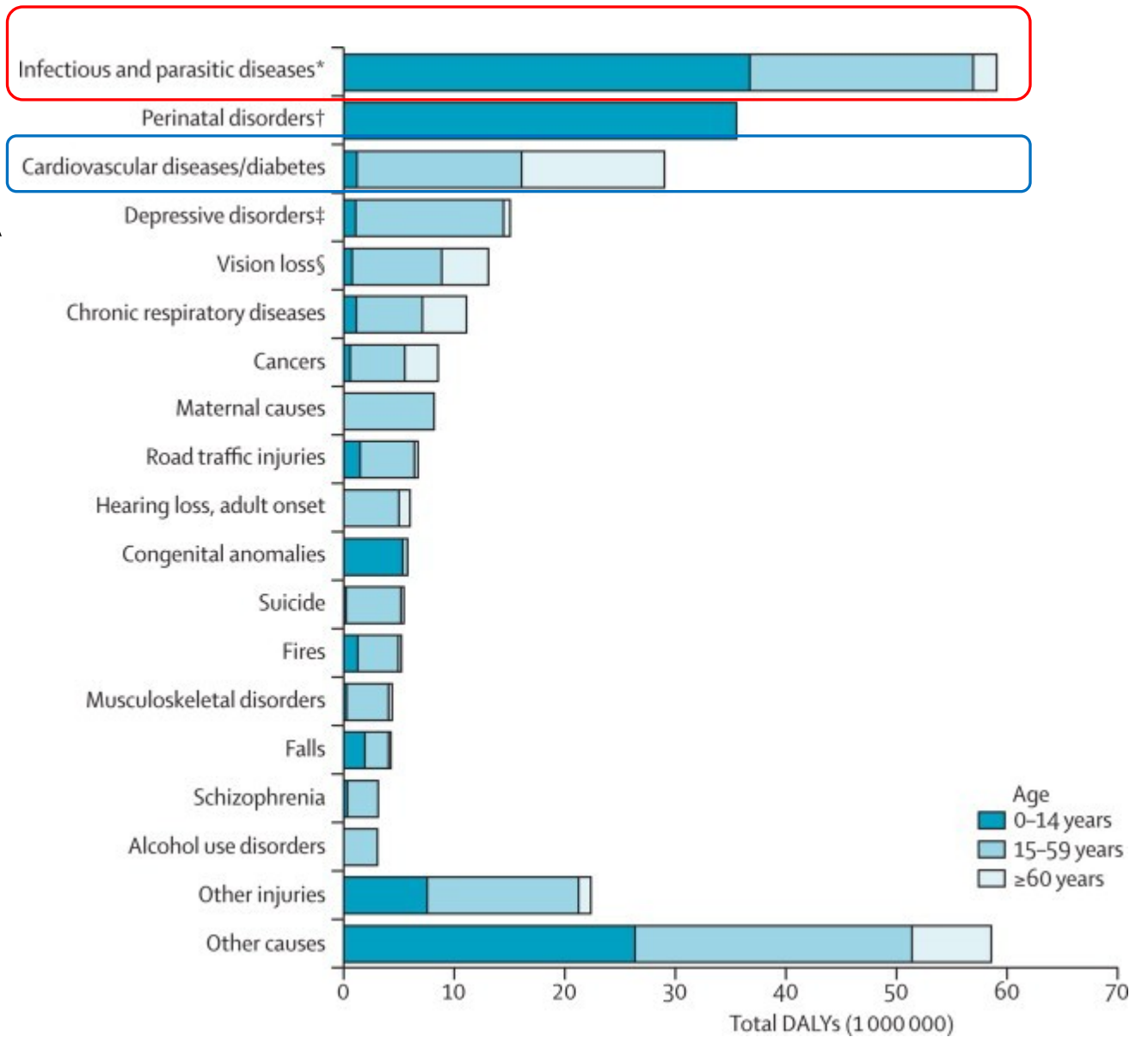
- (a) Communicable Diseases
- (b) Non-Communicable Diseases
- (c) Accidents/Self injuries

National Burden Estimates of healthy life lost in India, 2017: an analysis using direct mortality data and indirect disability data

Geetha R Menon, Lucky Singh, Palak Sharma, Priyanka Yadav, Shweta Sharma, Shrikant Kalaskar, Harpreet Singh, Srividya Adinarayanan, Vasna Joshua, Vaitheeswaran Kulothungan, Jeetendra Yadav, Leah K Watson, Shaza A Fadel, Wilson Suraweera, M Vishnu Vardhana Rao, R S Dhaliwal, Rehana Begum, Prabha Sati, Dean T Jamison, Prabhat Jha



DALY- INDIA



Patel, V., et al. (2011). Chronic diseases and injuries in India. *The Lancet*, 377(9763), 413-428.

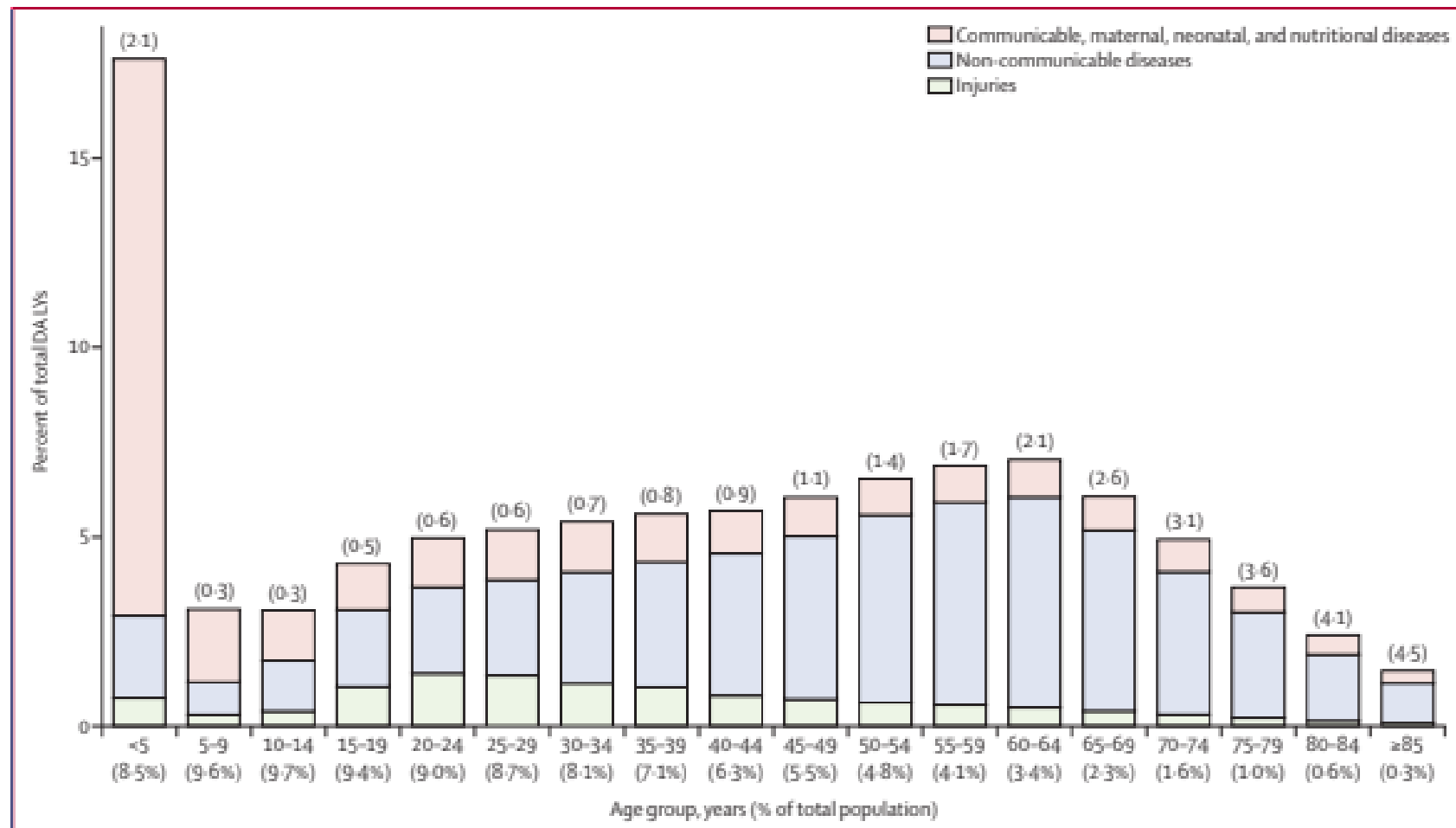
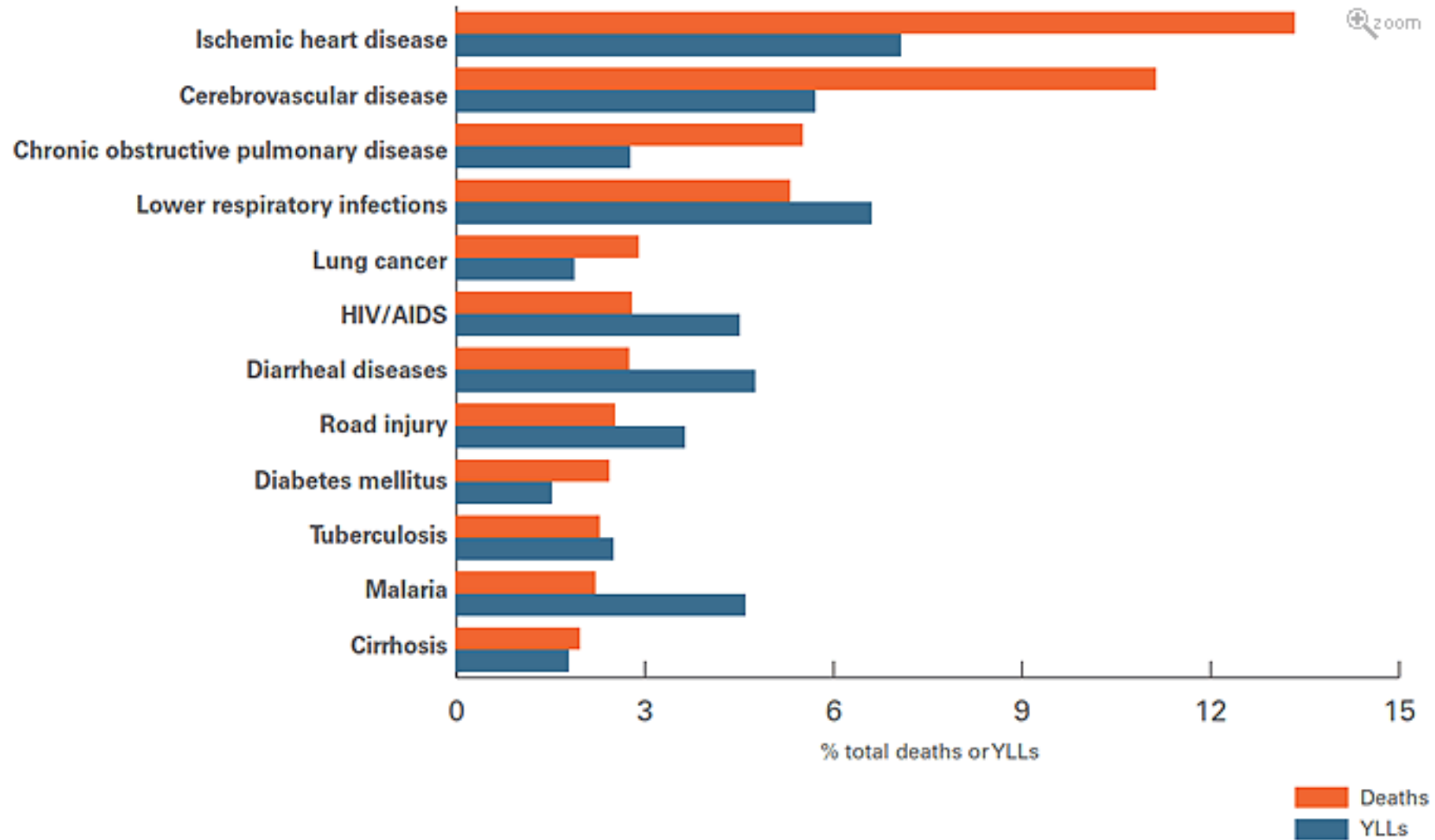


Figure 5: Percent of total DALYs by age groups in India, 2016

The number on top of each vertical bar is the ratio of percent DALYs to population for that age group. DALY=disability-adjusted life-year.

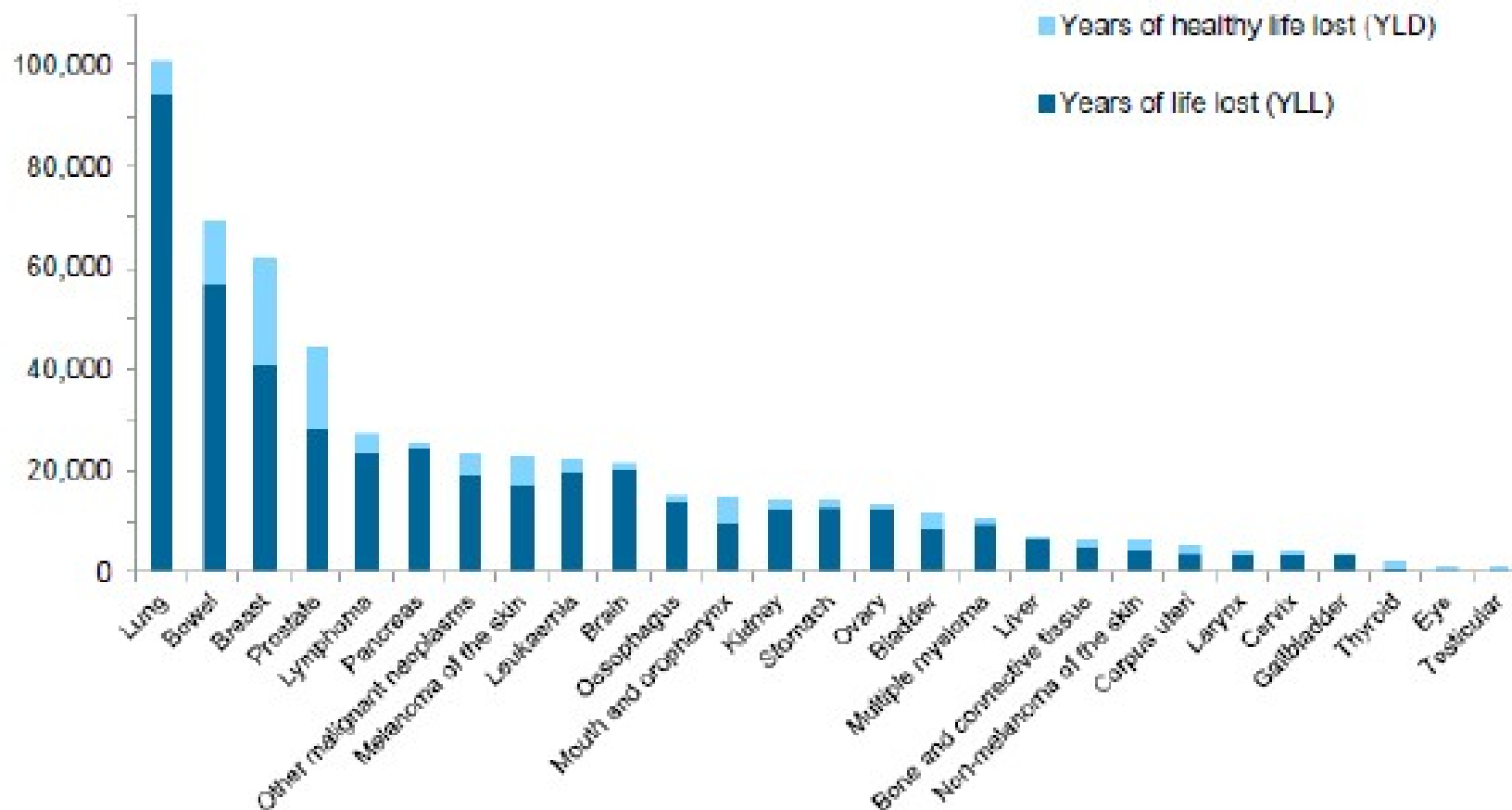
Nations within a nation: variations in epidemiological transition across the states of India, 1990–2016 in the Global Burden of Disease Study

YEARS OF LIFE LOST



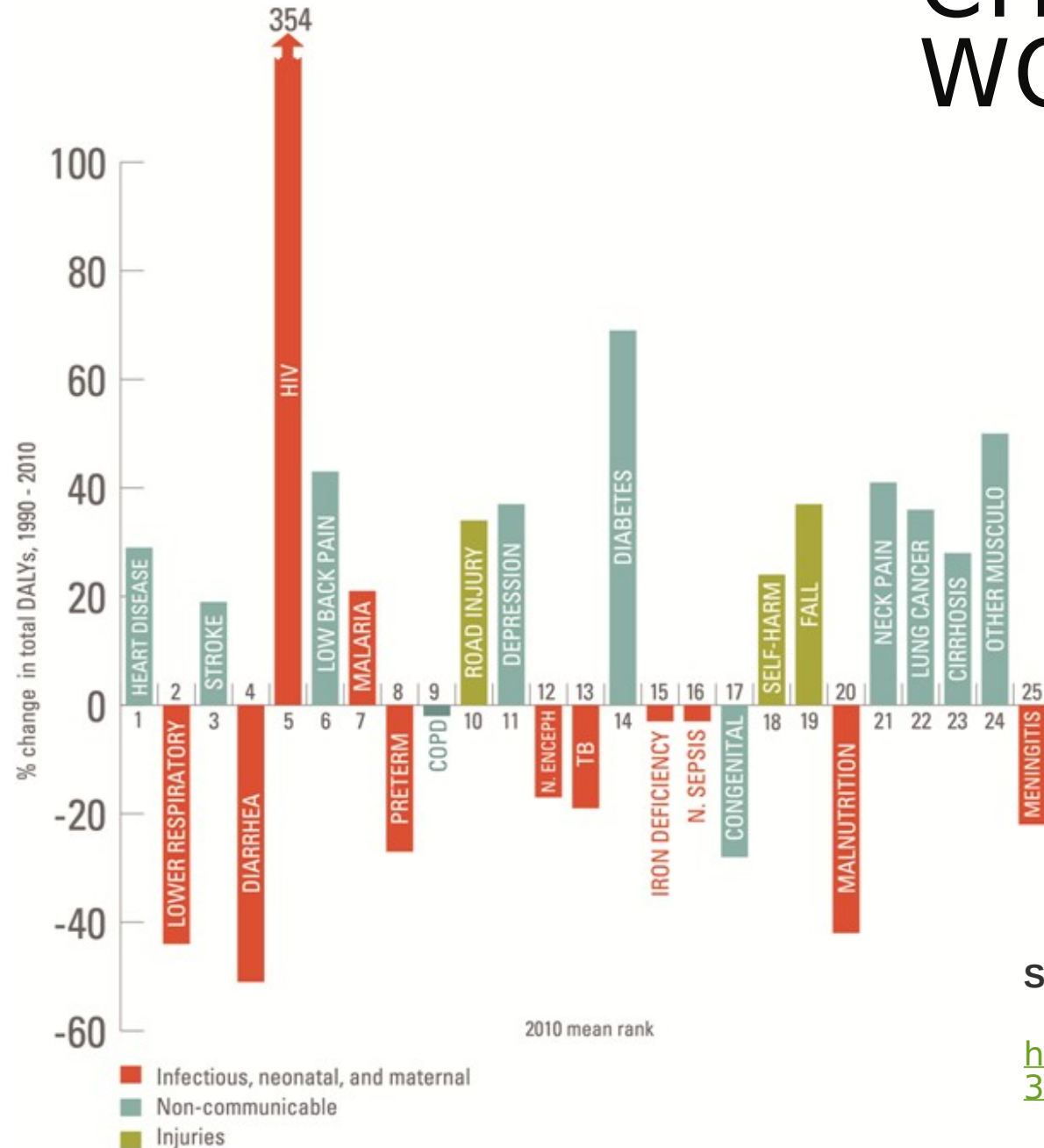
BURDEN OF CANCER

Disability-adjusted life years (DALYs)



Burden of disease due to cancer
Cancer in Australia: an overview 2012

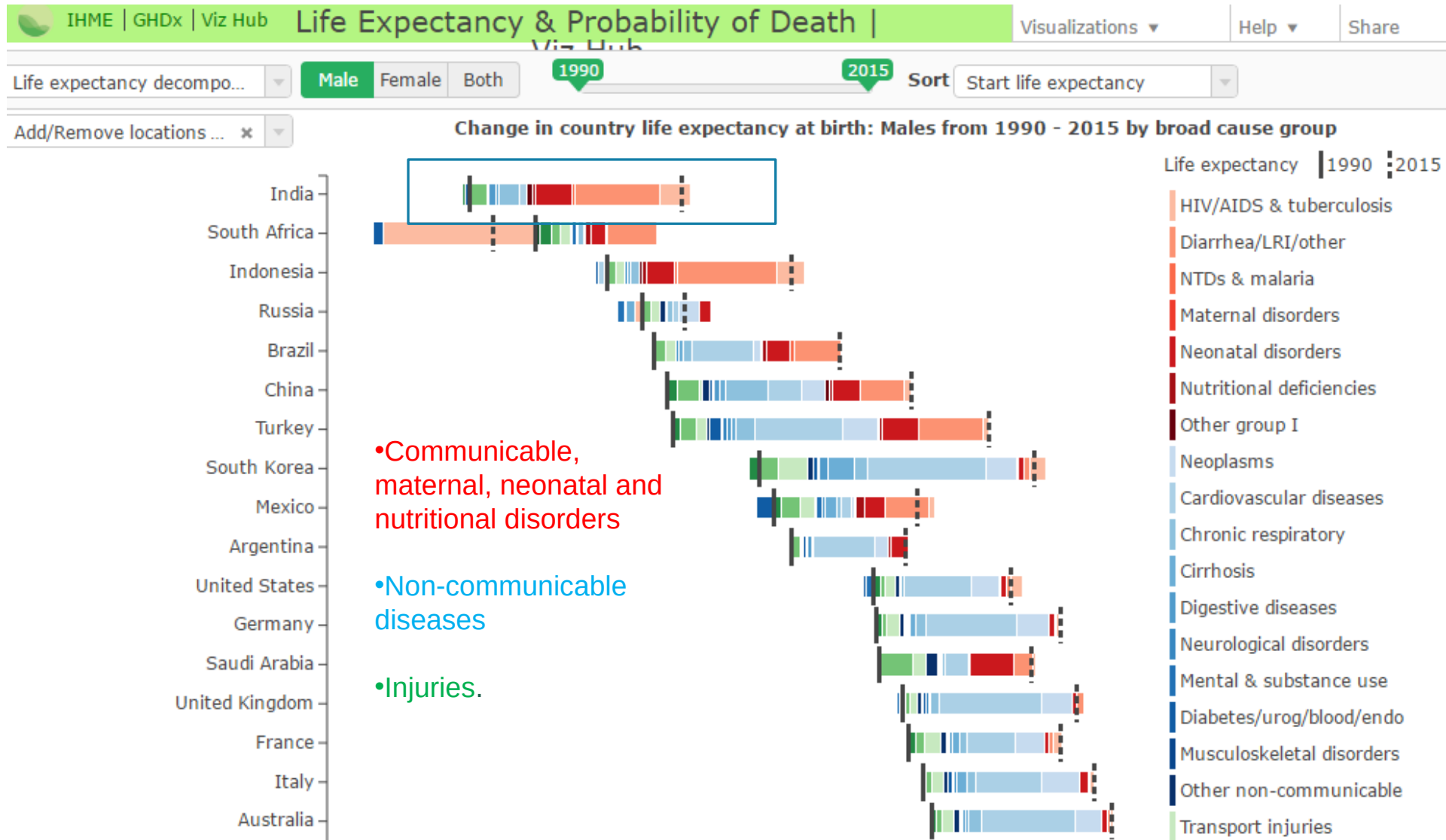
CHANGE IN DALY- WORLD 1990-2010



Source: Institute for Health Metrics and Evaluation (2013).

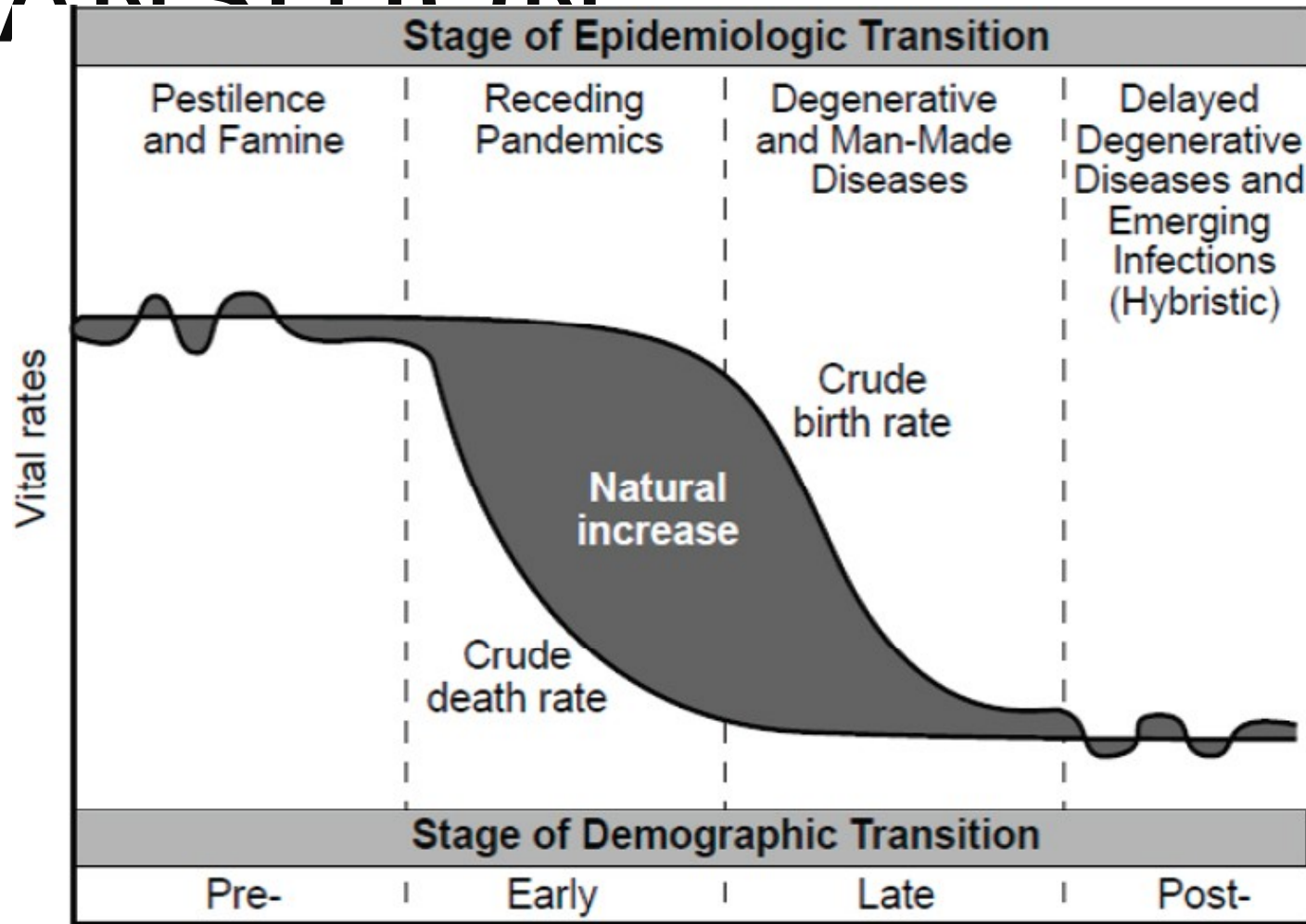
http://papp.iussp.org/sessions/papp104_s05/PAPP104_s05_050_030.html

GLOBAL BURDEN OF DISEASE



Institute for Health Metrics and Evaluation (IHME). Life Expectancy & Probability of Death Data Visualization. Seattle, WA: IHME, University of Washington, 2016.
<http://www.thelancet.com/lancet/visualisations/life-expectancy>

EPIDEMIOLOGICAL TRANSITION



[Demographic and epidemiological transition framework.](#)

SOURCES: As presented by Tim Evans on June 13, 2018. From [Rockett, 1999.](#)

<https://www.nap.edu/read/25276/chapter/3#6>

EPIDEMIOLOGICAL TRANSITION

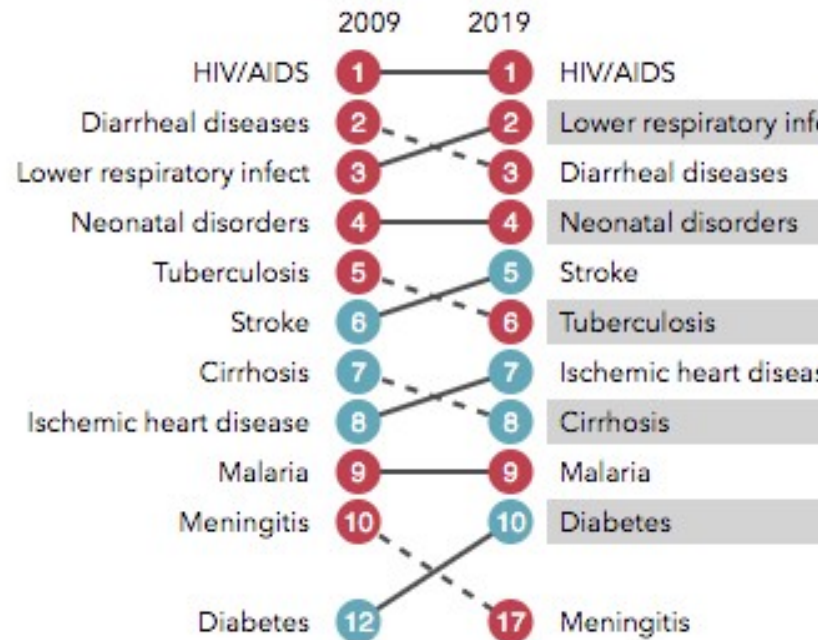
The mortality transition in more developed countries involved a shift in the **major causes of death from such communicable diseases as measles, influenza, and smallpox, to chronic and degenerative diseases such as heart disease, cancer, and emphysema.** This fundamental change reflected a broader change in health status and health threats that is described as the epidemiologic transition.¹⁷

- ***The Age of Pestilence and Famine*** when mortality is high and fluctuating, thus precluding sustained population growth. In this stage the average life expectancy at birth is low and variable, vacillating between 20 and 40 years.
- ***The Age of Receding Pandemics*** when mortality declines progressively; and the rate of decline accelerates as epidemic peaks become less frequent or disappear. The average life expectancy at birth increases steadily from about 30 to about 50 years. Population growth is sustained and begins to describe an exponential curve.
- ***The Age of Degenerative and Man-Made Diseases*** when mortality continues to decline and eventually approaches stability at a relatively low level. The average life expectancy at birth rises gradually until it exceeds 50 years. It is during this stage that fertility becomes the crucial factor in population growth.

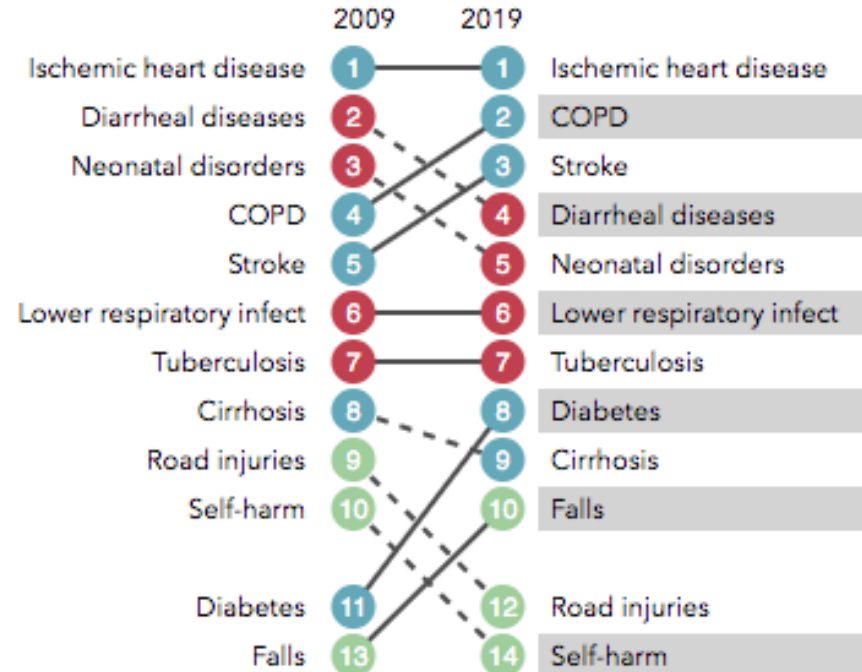
What causes the most deaths?

- Communicable, maternal, neonatal, and nutritional diseases
- Non-communicable diseases
- Injuries

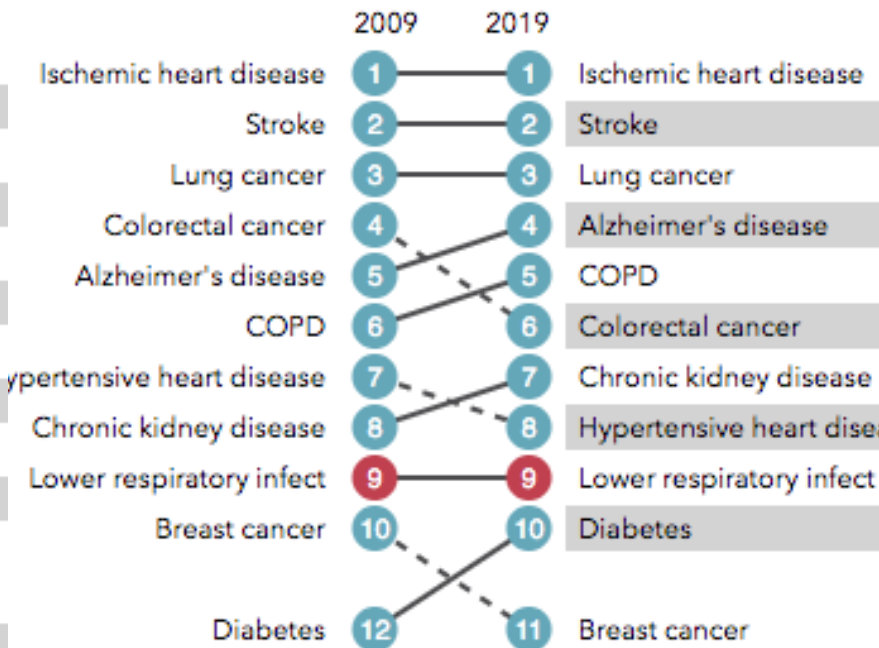
KENYA



INDIA



GERMANY



COPD Causes and Risk Factors



asthma



pollution



smoking

40+



age



chemical exposure

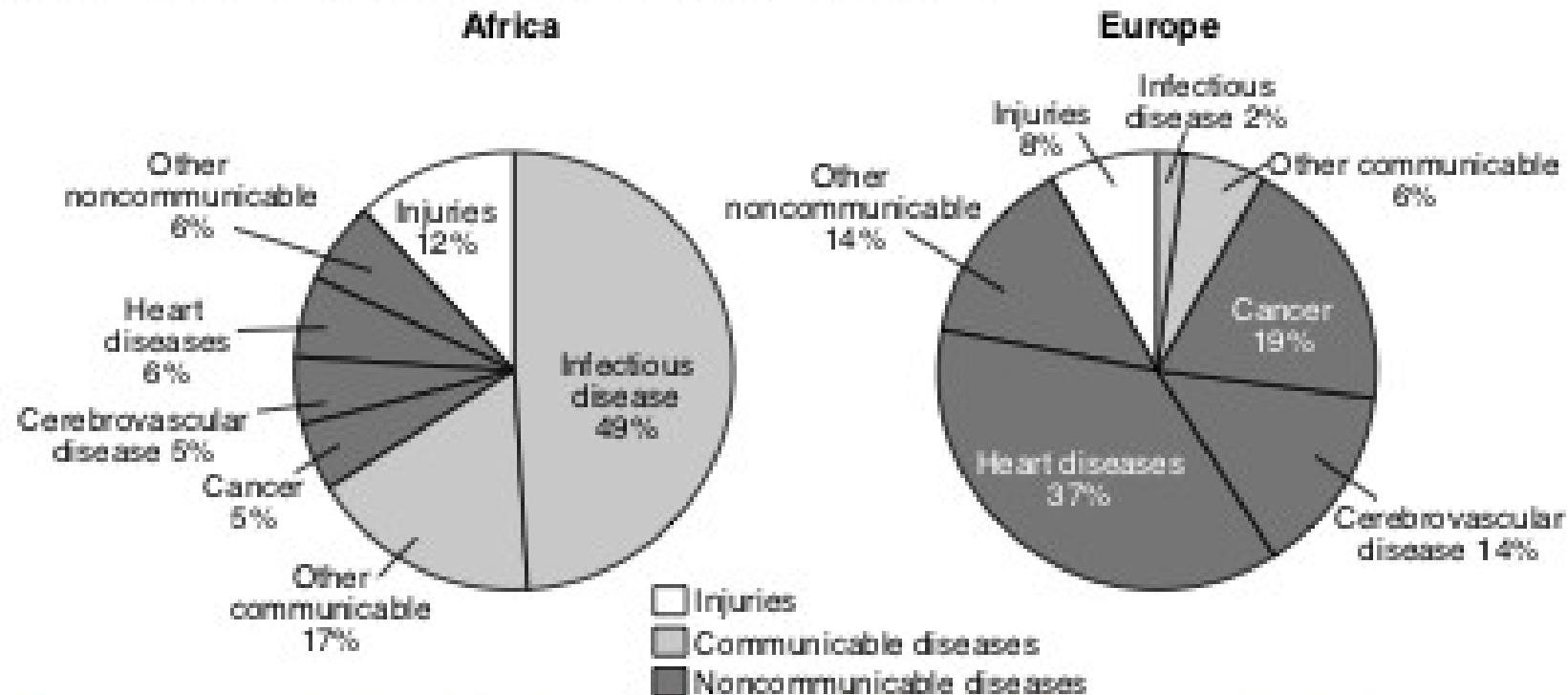


AAT deficiency



chronic bronchitis

Figure 4
Leading Causes of Death in Africa and Europe, 1998



Note: Infectious diseases causing most deaths include tuberculosis, HIV/AIDS, diarrheal diseases, malaria, and measles and other childhood diseases. Other communicable diseases include such diseases as pneumonia, bronchitis, and other lower respiratory diseases. Other noncommunicable diseases include such diseases as diabetes, mental disorders, chronic lung diseases, and cirrhosis of the liver. Injuries include intentional as well as unintentional injuries.

Source: WHO, *World Health Report, 1999* (1999): 102-3.

Scott C. Ratzan, Gary L. Filer- man, and John W. LeSar, "Attaining Global Health: Challenges and Opportunities," *Population Bulletin*, vol. 55, no. 1

AntiMicrobial resistance (GRAM)

Subscribe

OCTOBER 15, 2020

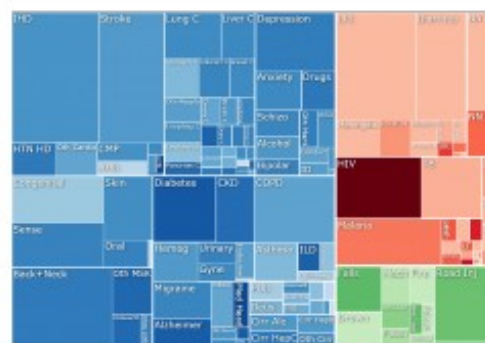
Data Visualization

Analyze updated data about the world's health levels and trends from 1990 to 2019 in this interactive tool using estimates from the Global Burden of Disease (GBD) study. Compare causes, injuries, and risks within a country, compare countries with regions or the world, and explore patterns and trends by country, age, and gender.

OCTOBER 15, 2020

Data Visualization

How do input data become GBD estimates? Walk through the estimation process for mortality trends for children and adults for 204 countries and territories using the Mortality visualization. See source and comparative data and step through the stages in the estimation process to reveal the final mortality estimates from 1950 to 2019.





IHME

Measuring what matters

- Home
- Results
- News & Events
- Projects
- Get Involved
- About

India

< Back

Language

English

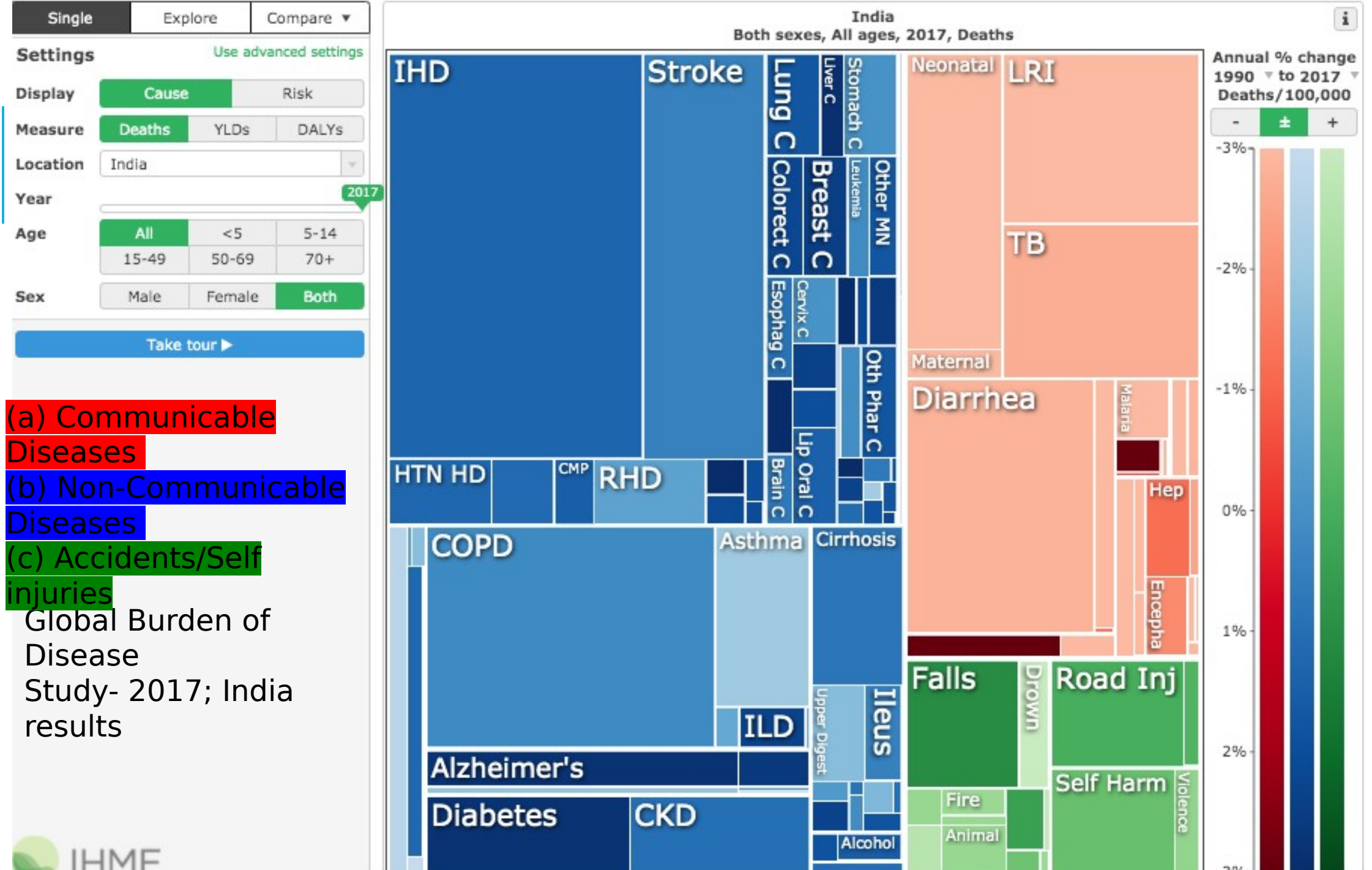
submit



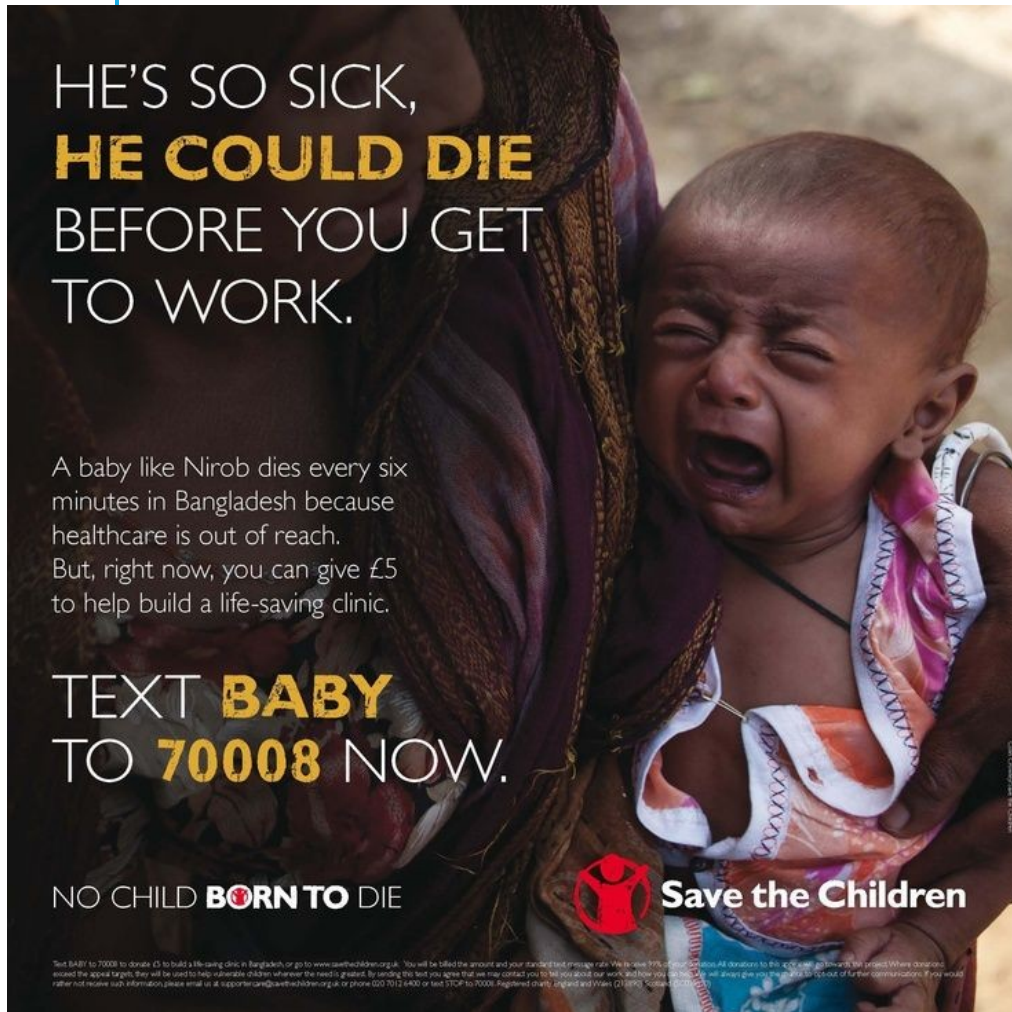
Print

2019 population	2019 fertility rate	2019 educational attainment (years)
1.4B	2.0	6.7

How is the population forecasted to change?



DEATH, EMOTIONS AND STATISTICS



HE'S SO SICK,
HE COULD DIE
BEFORE YOU GET
TO WORK.

A baby like Nirob dies every six minutes in Bangladesh because healthcare is out of reach. But, right now, you can give £5 to help build a life-saving clinic.

TEXT **BABY**
TO **70008** NOW.

NO CHILD **BORN TO DIE**

 **Save the Children**

Text 74881 to 70008 to donate £5 to build a life-saving clinic in Bangladesh or go to www.savethechildren.org.uk. You will be billed the amount and your standard text message rate. We'll send you a 70008 receipt. All donations to the UK's largest children's charity go to the ground where they are needed most. If you agree that we may contact you to tell you about our work, and how you can help, we will always give you the right to opt out of further communication. If you would rather not receive such information, please email us at supporters@savethechildren.org.uk or phone 020 7012 6400 or text STOP to 70008. Registered charity, England and Wales (218190). Scotland 020190.



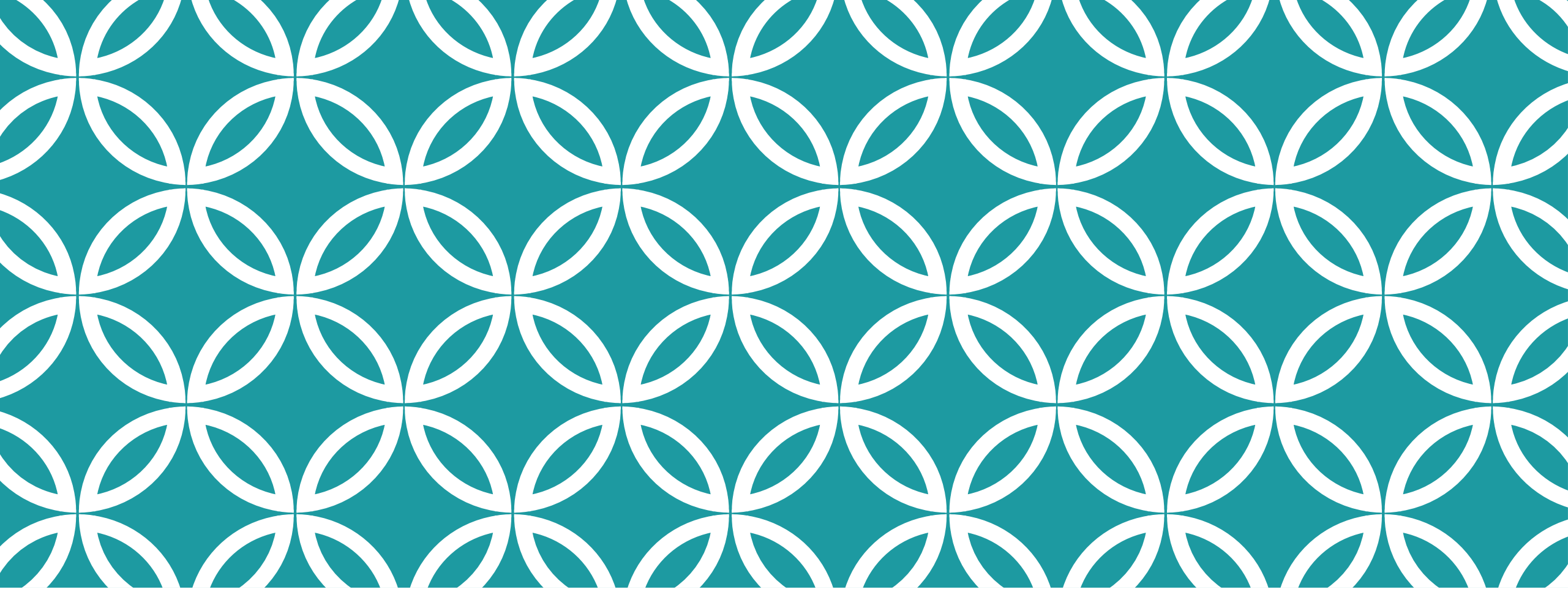
My friend saw her bus
She didn't see the car
Stop. Think. Live.

MAYOR OF LONDON  Transport for London 



**HE'S STARVING.
WE'RE NOT.
IT'S TIME TO SHARE >>**

unicef 



PUBLIC HEALTH SYSTEM IN INDIA |

HEALTH CARE IN INDIA

Main Systems:

- Western : **Allopathy (Evidence based medicine)** , Homeopathy
- Indian: Ayurveda , Unani , Siddha.. (Mystic healing ? 😊)--
AYUSH

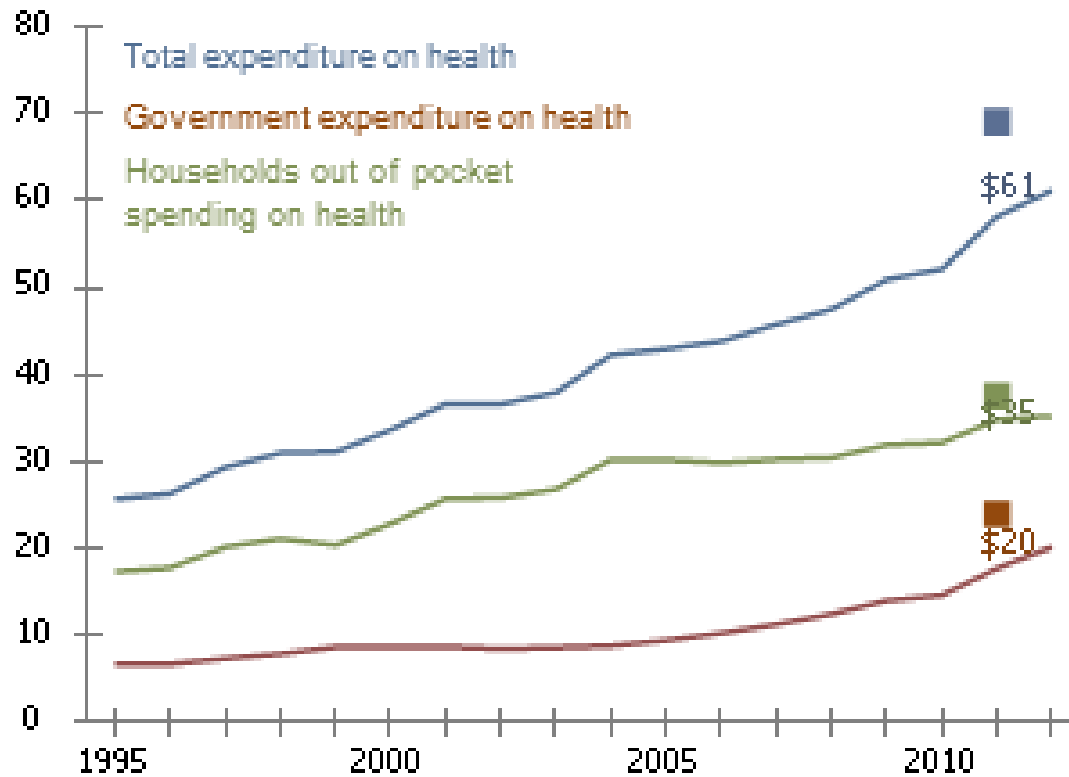
Ownership :

- Public : Sub-center, Primary health center (PHC), Community health center (CHC), District hospitals, ESICs
- **Private:**
 - **For profit**
 - Not for profit (run by NGOs , trusts)

HEALTH EXPENDITURE

Per capita expenditure in US\$
(constant 2012 US\$)

■ 2012 average of SEAR Low-Mid
income countries



World
Bank,

2019 – Public. 1.29%
GDP

Most spending on curative
rather than preventive health

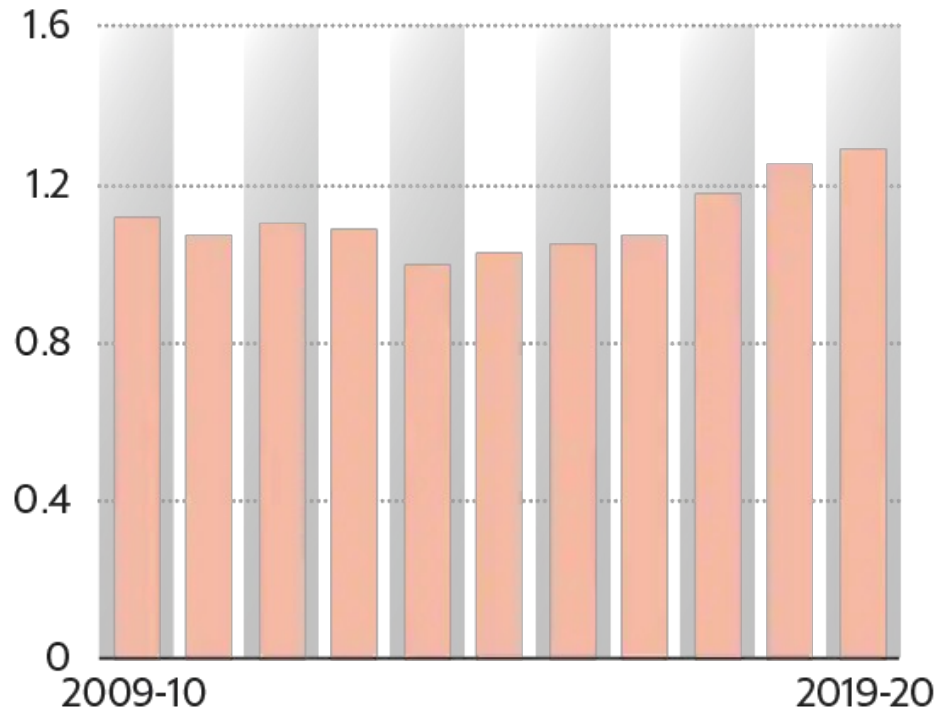
Doctor to pupil:
One doctor for every 1,445 Indians
(wide inequality between states, across
districts
Tribal , Rural,, and Urban areas)

(National Health Profile, 2019)

Health a low priority

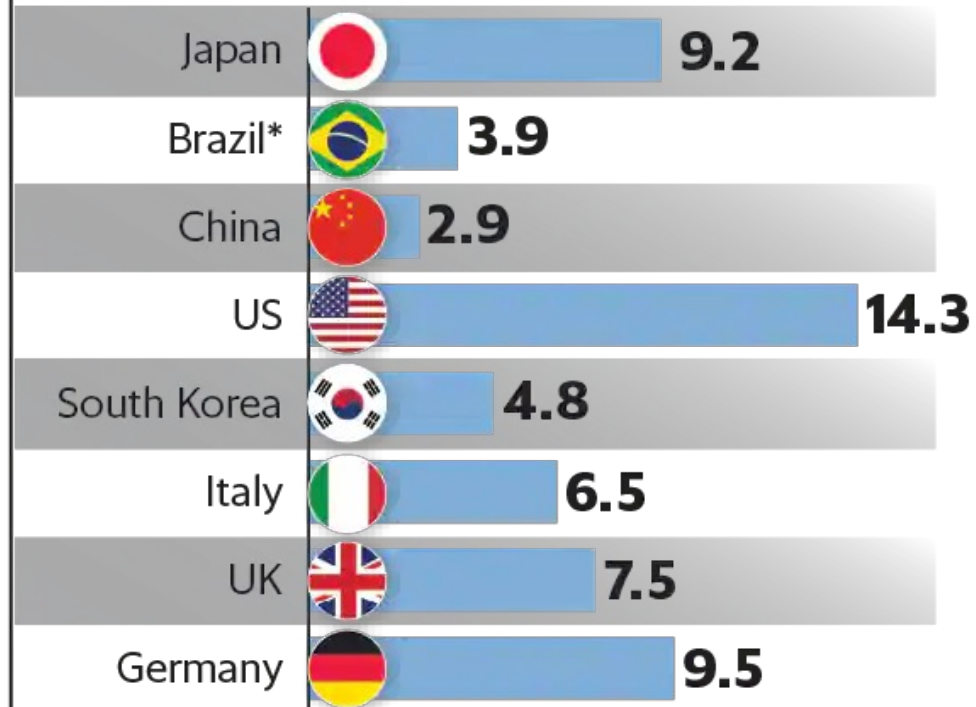
India's public health expenditure was just 1.29% of GDP in 2019-20. In 2018 too, the country lagged behind BRICs peers as well as developed nations.

India's (centre plus states) public expenditure on health (as % of GDP)



Source: National Health Profile 2019, Government of India; CARE Ratings' calculations

Public expenditure on health in 2018 (as % of GDP)



*For 2017

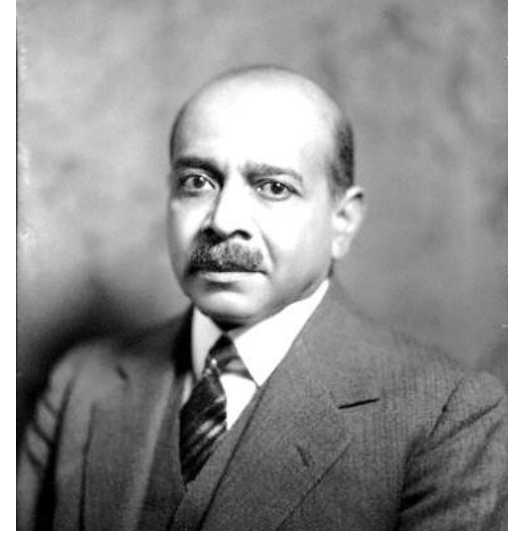
Source: OECD, Conta-Satélite de Saúde

MILESTONES IN INDIAN HEALTH POLICIES

Bhore committee: 1946 (Joseph Bhore)

"The existing state of public health in the country is so unsatisfactory that any attempt to improve the present position must necessarily involve administrative measures of such magnitude as may well seem to be out of all proportion to what has been conceived and accomplished in the past.."

*"It were possible to evaluate the loss, which **this country annually suffers through the avoidable waste of valuable human material and the lowering of human efficiency through malnutrition and preventable morbidity**, we feel that the result would be so startling that the whole country would be aroused and would not rest until a radical change had been brought about "*



Bhore, Joseph, 1946 : Report of the Health Survey and Development Committee, Volume I to IV, Govt. of India, Delhi

HEALTH UNDER CONSTITUTION

Health- Primarily a **state subject** (7th schedule)

Public health and sanitation; hospitals and dispensaries. Relief of the disabled and unemployable

Central List: *Regulation of labour and safety in mines and oilfields, Port quarantine, including hospitals connected therewith*

Concurrent List (Joint responsibility of state and center):
Mental health, Social Security, Social Insurance, Statistics, Medical Education etc.



HEALTH CARE DELIVERY



Health Infrastructure in India's Urban & Rural Communities- Shamika Ravi
<https://www.youtube.com/watch?v=C2AHSZ8Hgsk&t=125s>

PUBLIC HEALTH DELIVERY SYSTEM

Primary health care

- Sub-Centers , Primary Health Centers (PHC)

Hospitals, Health centers

- Community Health Centers (CHC)
- Rural hospitals, District hospital
- Specialist hospitals, Teaching hospitals

Health Insurances schemes

- Employees state insurance (ESIC), Central Govt. Health Scheme
- Rashtriya Swastya Bima Yojna (RSBY)

District Hospitals

CHC

80,000-100,000 people

PHC

20,000-30,000 people

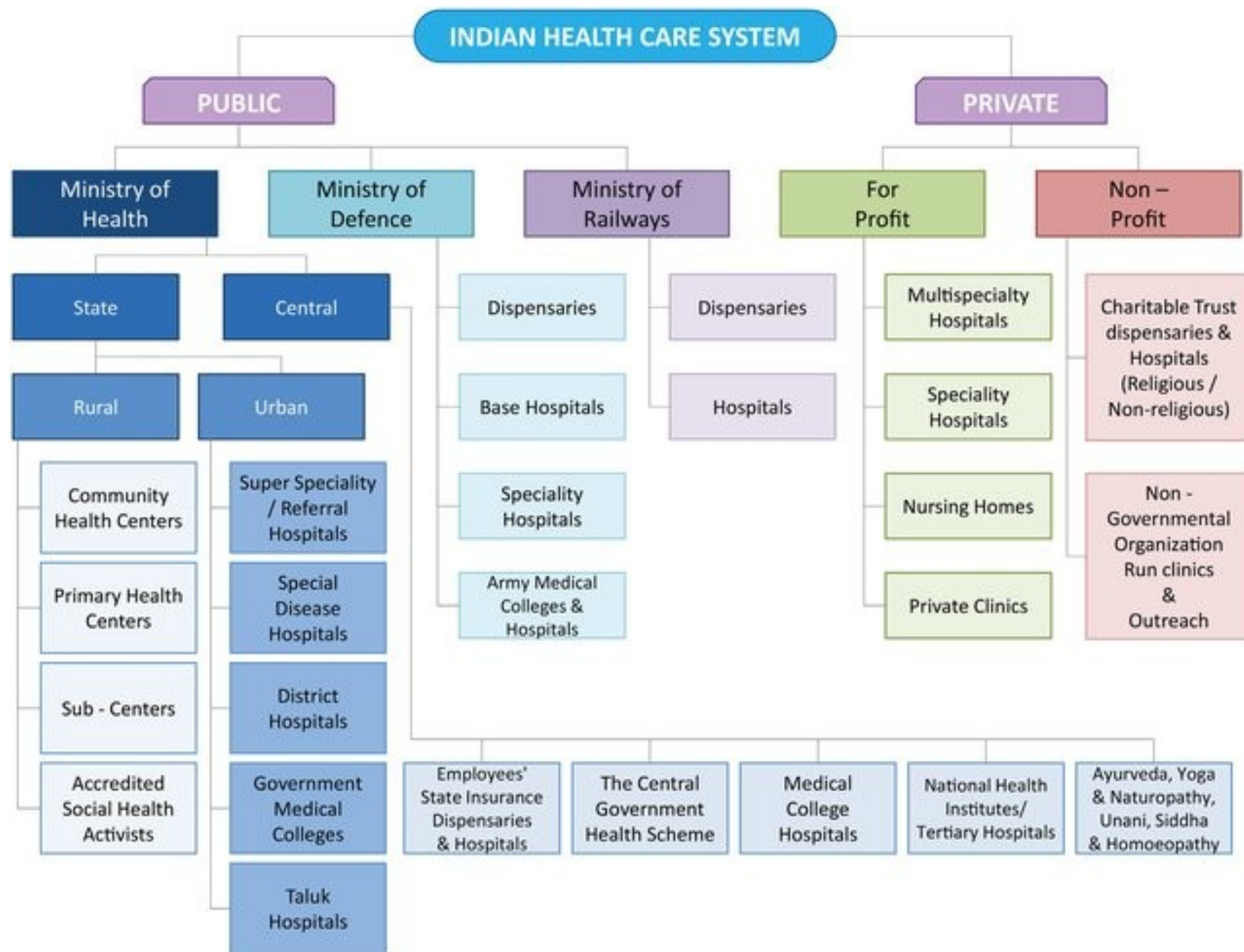
Sub-center

3000-5000 people

Village

Anganwadi Worker
(ICDS)

one per 1000



* Image adapted from Ramadas S. Is Indian Health Care System better than US ? A Sample Study ... [Internet]. LinkedIn (2014). Available from: <https://www.linkedin.com/pulse/20140729152938-63974960-is-indian-health-care-system-better-than-us-a-sample-study>

An overview of Indian healthcare system. The Challenges and Recommendations for Gestational Diabetes Mellitus Care in India: A Review March 2017 Frontiers in

SUB-CENTER

One center per 3000-5000 population

Two **health workers** (male and female)

Forms a link between village health workers and PHC

Duties:

- ▣ Antenatal, natal, postnatal care
- ▣ **family planning and counselling**
- ▣ treatment of common illnesses like respiratory tract infections, Diarrhoea, fever, worm infestation
- ▣ prevention of malnutrition
- ▣ implementation of various national health program



PRIMARY HEALTH CENTER

One for 20,000 to 30,000 population

Basic unit of health care delivery

Medical officer, pharmacist, nurse etc

Functions:

- ▢ Basic medical care
- ▢ Maternal and Child Care
- ▢ Prevention of epidemics
- ▢ Water supply and sanitation!
- ▢ Collection and reporting of vital statistics
- ▢ Implementation of national health programs



COMMUNITY HEALTH CENTER

- One CHC 80,000 to 100,000 population
- Secondary of health care delivery
- 30 bed facility with specialists
- Functions:
 - Out patient and inpatient departments:
General, Medicine, Surgery, OBG
Pediatrics, Dental and AYUSH services
 - Emergency Services
 - Laboratory and Diagnostic facilities
 - Implement National health program (NRHM, JSY etc)



Image: Institute of public health , Bangalore

PUBLIC HEALTH INSURANCE IN INDIA

Employees State Insurance Scheme, (ESIS),

- ▮ Covers ALL ORGANIZED sector employees earning less than **Rs 15,000 per month (2011- 12)**
- ▮ Contribution by Employer and Employee (minimal)
- ▮ Free Cashless service at all ESI hospitals
- ▮ 15.5 million employees, 65.5 million beneficiaries (2011-12)

Central Government Health Scheme (CHGS)

- ▮ Central Govt. employees and pensioners and their dependents residing in CGHS covered cities. 17 cities in India

AYUSHMAN BHARAT

Ayushman Bharat Yojana aims to cover more than 50 crore Indian citizens.

- It is designed especially for the economically weaker sections of the country.

The PMJAY was launched in September 2018 providing health insurance coverage of a **maximum sum insured amount of Rs.5 lakh**.

The government health insurance scheme covers most of the **medical treatment costs, medicines, diagnostics and pre-hospitalisation expenses**.

- Additionally, the scheme offers cashless hospitalisation services through the Ayushman Bharat Yojana e-card which you can use to get healthcare services at any of the empanelled hospitals across the country.
- Beneficiaries of the scheme can avail hospitalisation for necessary treatment by showing their PMJAY e-card.

General OPD not covered, Cosmetic surgeries, Fertility treatments, Individual diagnostics, Organ transplant.

DISEASE SPECIFIC PROGRAMS

National Malaria Control Program – 1953

□ Vector Borne : Kala Azar, JE, Filaria, Dengue

National Leprosy Eradication Programme- 1955

National Iodine Deficiency Disorders Control Programme

National Tuberculosis Control Program (1962, 1992)

National Aids Control Programme (1992, 2002)

INTEGRATED CHILD DEVELOPMENT SCHEME(ICDS)

Launched in 1975, ICDS is World's largest early childhood development program

Anganwadi centers through out India

Objectives :

- Improve the nutritional and health status of children in the age-group 0-6 years
- Reduce the incidence of mortality, morbidity, malnutrition
- to enhance the capability of the mother to look after the normal health and nutrition

Services : supplementary nutrition, immunization, health check-up, referral services, pre-school non-formal education and nutrition & health education

COMMENT ON FUNCTIONING OF HEALTH SERVICES

While the Indian Public health system is , on paper, an ideal health care system for a developing country, it's all but defunct in practice, due to the lack of interest of both providers and putative patients

- Abhijit Banerjee, Ester Duflo, “Improving health care delivery in India”, Angus Deaton Festschrift , Princeton

(Based on studies in rural Udaipur, Rajasthan)

Also discussed in the book “Poor Economics”

